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Chained Indices Unchained: Structural Transformation and the Welfare Foundations of Income Growth Measurement*

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Abstract

This paper examines how to measure welfare-relevant growth in economies undergoing structural transformation, where persistent changes in relative prices and expenditure patterns can lead to aggregation biases in standard output measures. We address this issue within a continuous-time dynamic general equilibrium model featuring non-homothetic preferences, sector-specific productivity trends, and investment-specific technical change. We compare two welfare-based measures of real income growth: a current-base equivalent variation measure and a chained Fisher–Shell index. The chained Fisher–Shell index coincides with the chained Divisia index, providing strong theoretical foundations for national accounting methodology and requiring the growth rates and nominal expenditure shares of major expenditure components as sufficient statistics for a welfare-based output index. In contrast, the current-base index is more informationally demanding, requires systematic revisions, and introduces base-dependent biases that distort the evaluation of past growth. Our quantitative analysis reveals that these biases are substantial, increase over time, and conflict with key stylized facts in the structural transformation literature.

KEYWORDS: Structural transformation, Non-homothetic preferences, Investment-specific technical change, Chained quantity indices, GDP measurement, Equivalent variation, Divisia index, Fisher–ideal index, Fisher–Shell index

JEL CLASSIFICATION NUMBERS: C43, E01, E13, O11, O14, O41, O47.

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1 Introduction

The Kaldor facts ([Kaldor, 1961](#)) have long provided the empirical foundation for balanced growth models in macroeconomics. These models emphasize the constancy of output and capital growth rates, factor income shares, and the capital-output ratio. Yet, driven by structural transformation and embodied technological progress, sustained changes in expenditure patterns, relative prices, and sectoral composition raise questions about the universality of balanced growth and the adequacy of standard output measurement.¹ When growth is unbalanced, persistent shifts in relative prices and consumption patterns may lead to aggregation biases—particularly when output is measured using traditional fixed-base quantity indices. These biases can result in discrepancies between measured real GDP and welfare-relevant output. This observation motivates the central question of this paper: *how should we measure welfare-relevant output growth in economies undergoing structural reallocation in consumption, from goods to services, and experiencing investment-specific technical change that persistently lowers the relative price of capital?*

The analysis focuses on the chained Divisia index, which provides a theoretical benchmark for the chained Fisher–ideal index commonly used in national accounts to correct the measurement biases introduced by fixed-base aggregation. We examine how the measurement of real income through such indices affects the evaluation of welfare gains in economies undergoing structural change and embodied technical progress.

Recent theoretical work has offered contrasting interpretations of how chained indices relate to welfare when preferences evolve over time. We compare two such approaches: [Baqae and Burstein \(2022\)](#) and [Durán and Licandro \(2025\)](#).² Both build on the Fisher–Shell principle ([Fisher and Shell, 1968](#)), which states that welfare comparisons under time-varying preferences require a common preference order, with current preferences serving as a natural baseline.

While both frameworks adopt this principle, they differ in how it is implemented.

¹See the related literature section below for further discussion of the unbalanced growth properties of structural transformation and embodied technical progress models.

²See also the foundational paper by [Licandro et al. \(2002\)](#), which proposes the application of the Fisher–Shell principle to the measurement of output growth, and the discrete-time application in [Duernecker et al. \(2021\)](#).

Baqae and Burstein (2022) construct an equivalent variation measure using current preferences to evaluate the entire history of income growth. In the following, we refer to it as a *current-base Fisher–Shell quantity index*. This approach requires stability and homotheticity of preferences to ensure that chained Divisia indices are consistent with welfare. Absent these conditions, the chained Divisia index may diverge from their equivalent variation measure. In contrast, Durán and Licandro (2025) apply the Fisher–Shell principle in a local, dynamic setting, constructing an equivalent variation measure by using preferences at each point in time to evaluate marginal welfare gains, which are then integrated along the growth path. They refer to it as a *chained Fisher–Shell quantity index* and show that in continuous-time dynamic general equilibrium models it coincides with the chained Divisia index. Their approach does not require stable or homothetic preferences, and it allows the index to adapt flexibly to evolving income, preferences, and prices.

We quantitatively evaluate these two alternative welfare-based measures of real income growth within a continuous-time structural transformation framework inspired by Herrendorf et al. (2021). In particular, we assess how the use of either the equivalent variation measure of Baqae and Burstein (2022) or the chained Fisher–Shell quantity index of Durán and Licandro (2025) affects measured growth in an economy undergoing structural change. The model integrates the two key drivers of unbalanced growth, structural transformation and investment-specific technological change (ISTC), and embeds non-homothetic preferences, sector-specific productivity trends, and a flexible technology for the production of investment goods. This makes it ideally suited to study how changes in preferences, technology, and prices shape both real income dynamics and their measurement.

Our analysis shows that the chained Fisher–Shell index –by continuously updating preferences and prices– yields a welfare-consistent measure of real GDP growth, exactly equal to a chained Divisia index. This measure is invariant to the choice of reference time and accurately tracks balanced and unbalanced growth properties of the economy. In contrast, the current-base index systematically underestimates welfare gains along the structural transformation path relative to the Divisia index, while a past-base index overestimates them. As time passes and past economic progress is repeatedly re-evaluated through successive current-base Fisher–Shell measures, the perceived gains from growth appear to shrink.

To see the intuition, consider the US structural transformation model that is the focus of our quantitative analysis. When we evaluate past income growth using today’s preferences (a current-base index), we overstate the historical share of consumption expenditure in GDP because, under ISTC, today’s preferences value consumption more relative to investment than past preferences did. Since consumption grows more slowly than investment along the structural transformation path, overstating the past consumption share leads measured real income growth to be understated. Conversely, when we evaluate past income growth using past preferences, we understate the historical share of consumption in GDP. Understating the past consumption share –the slower-growing component– causes measured real income growth to be overstated. The chained Divisia index avoids both biases by continuously updating preferences and prices along the growth path, delivering a welfare-consistent measure of real income growth.

This result underscores the central role of measurement in the analysis of structural transformation. Although chained indices such as the Fisher–ideal index are widely used in statistical practice –often for pragmatic reasons, such as limiting revisions when base years change– their deeper welfare-theoretic justification lies in their ability to accommodate evolving preferences and relative prices. From this perspective, the main contribution of this paper is to establish a conceptual foundation for welfare-based measurement in dynamic environments with evolving preferences and prices. This welfare-theoretic perspective highlights the essential role of chained indices in accurately capturing welfare-based output growth in dynamic economies.

To assess whether these findings depend on the particular PIGL specification, we replicate the quantitative analysis using the non-homothetic CES preferences proposed by [Comin et al. \(2021\)](#). This alternative specification separates the intertemporal elasticity of substitution from the parameters governing income effects across consumption categories and fits the observed structural transformation as well as the baseline model. The results remain essentially unchanged: the current-base Fisher–Shell index understates welfare growth relative to the Divisia index, whereas the reference-base index overstates it. Moreover, the quantitative discrepancies between the fixed-base and chained measures are similar to those obtained under PIGL preferences. The measurement results are therefore a general consequence of evaluating growth under non-homothetic preferences at different reference points, rather than an artefact of the particular preference specification adopted in the baseline model.

In addition to its theoretical implications, our approach yields a second major contribution: it establishes a direct and transparent link between model-based welfare measurement and observable data. Specifically, welfare-relevant output growth can be expressed as a chained Divisia index that depends only on the growth rates and nominal expenditure shares of the main components of aggregate expenditure. As shown by [Durán and Licandro \(2025\)](#), this index coincides with the chained Fisher–Shell index in continuous-time settings, and can be constructed from standard national accounts data without requiring auxiliary assumptions or data transformations.

This result implies that the key statistics commonly used in the structural transformation and investment-specific technical change literatures –namely, component-level growth rates and nominal expenditure shares– are sufficient to measure welfare-relevant output growth. Our analysis provides a theoretical guarantee that GDP measured using the Fisher–Shell index is consistent with welfare. This makes the index not only practically implementable using national accounts data, but also widely applicable across a broad class of structural change models, including those discussed in Footnote 3 below.

Related Literature. The broader context for our analysis lies in two literatures concerned with the dynamics of unbalanced growth: the structural reallocation of economic activity and the role of embodied technological change.

The literature on structural transformation has identified persistent shifts in sectoral labor shares, consumption patterns, and productivity as key features of long-run development ([Kuznets, 1966](#); [Duarte and Restuccia, 2010](#)). A substantial body of theory attributes these patterns to mechanisms such as non-homothetic preferences ([Kongsamut et al., 2001](#)), differential productivity growth ([Ngai and Pissarides, 2007](#)), and capital deepening ([Acemoglu and Guerrieri, 2008](#)). These mechanisms are often combined in unified frameworks, such as [Herrendorf et al. \(2013\)](#), to jointly capture income and substitution effects.³ Among these contributions, [Boppart \(2014\)](#) proposes the price-independent generalised-linear (PIGL) class of preferences, while [Comin et al. \(2021\)](#) develops a non-homothetic CES specification, both allowing income and relative-price effects to jointly shape the process of structural transformation.

The literature on embodied technical progress documents the secular decline in the

³See also [Herrendorf et al. \(2014\)](#) for a survey, and alternative formulations in [Buera and Kaboski \(2009\)](#), [Boppart \(2014\)](#), [Alvarez-Cuadrado et al. \(2017\)](#), [Swiecki \(2017\)](#), [Comin et al. \(2021\)](#), and [Alder et al. \(2022\)](#).

relative price of investment goods as a result of investment-specific technological change (ISTC) ([Gordon, 1990](#); [Greenwood et al., 1997](#); [Boucekkine et al., 2003](#)). This empirical regularity has prompted important revisions in national accounts methodology, including the adoption of chained Fisher–ideal indices by the BEA in 1996 to better reflect relative price changes and expenditure shifts.⁴ While these indices are now standard practice, their welfare-theoretic properties have received less attention. Our paper contributes to this literature by examining whether such indices capture welfare-relevant growth in environments with structural reallocation and ISTC, drawing on the theory of index numbers ([Fisher, 1922](#); [Fisher and Shell, 1968](#); [Diewert, 1976, 1978](#); [Caves et al., 1982](#)).

Building on these insights, [Herrendorf et al. \(2021\)](#) propose a unified framework that integrates structural transformation and embodied technical progress into a single general equilibrium model. Their approach allows for both sector-specific productivity trends and investment-specific technological change, capturing the dual sources of unbalanced growth documented in the data. By embedding vintage capital and non-homothetic preferences within a three-sector model, they show how the simultaneous evolution of preferences, productivity, and relative prices jointly determine the sectoral composition of output, consumption and investment over time. This framework enables a richer understanding of long-run growth dynamics and highlights the importance of using consistent measurement tools, such as chained indices and quality-adjusted prices, to accurately track welfare-relevant changes in economic activity.

A growing recent literature revisits the measurement of welfare using microeconomic data and flexible preference structures. For example, [Atkin et al. \(2023\)](#) develop a method to estimate welfare and inequality when price data are incomplete by exploiting shifts in Engel curves across households, allowing equivalent and compensating variations to be recovered from expenditure survey data under quasi-separable preferences.

A closely related contribution is [Baqae and Burstein \(2022\)](#), who study welfare measurement in environments with non-homothetic preferences and taste shocks. They construct a money-metric measure of welfare based on an equivalent variation concept that evaluates consumption bundles using current-period preferences. Aggregating across households, they derive a macro welfare index that can be used to rank alternative production possibilities frontiers and to compare welfare across equilibria. In contrast, our approach evaluates welfare changes along the growth path by chaining preferences

⁴See [Parker and Triplett \(1996\)](#) and [Whelan \(2002\)](#).

over time, rather than fixing them at a single reference date. This allows welfare changes to be measured locally along the transition path. Moreover, because preferences admit a Bellman representation in the dynamic environment we study, welfare comparisons can be conducted directly at the micro level without the need to construct an aggregate welfare index. Building on this framework, [Baqaee et al. \(2023\)](#) propose a method to measure welfare by matching households across time in micro data, allowing welfare comparisons even when the cross-sectional distribution of consumption and preferences evolves. Related work by [Jaravel and Lashkari \(2023\)](#) develops nonparametric methods to measure consumer welfare growth under income-dependent preferences, highlighting how non-homothetic demand and product-level price changes shape welfare-relevant price indices.

While these contributions emphasize the measurement of welfare using detailed micro data and flexible demand systems, our paper focuses on the welfare interpretation of aggregate output indices commonly used in macroeconomic analysis and national accounts. In particular, we study whether chained quantity indices, such as the Divisia or Fisher indices, provide a welfare-consistent measure of output growth in dynamic environments with structural transformation and investment-specific technological change.

2 Structural Transformation Model

To compare the behavior of fixed-base Fisher–Shell indices with that of the chained Fisher–Shell index, this section adopts a continuous-time version of the structural transformation model developed by [Herrendorf et al. \(2021\)](#). Hereafter, we refer to it as the HRV structural transformation model. In a closed economy, the model features three sectors: goods, services, and investment, with non-homothetic preferences defined over consumption of goods and services. Both goods and services are also used in the production of investment, which is governed by a CES technology. This framework provides a rich environment for evaluating how fixed-base and chained true quantity indices capture welfare-relevant income growth along a balanced growth path with non-homothetic preferences, time-varying income elasticities, and a constant elasticity of substitution.

Description of technology. We assume that goods and services are produced by distinct sectors. Value added in each sector is generated using Cobb–Douglas production

technologies

$$Y_{j,t} = A_{j,t} K_{j,t}^\theta L_{j,t}^{1-\theta},$$

where $j, j \in \{g, s\}$, indexes the goods and services sectors, respectively. These production functions share a common capital intensity parameter θ , $\theta \in (0, 1)$, but differ in total factor productivity, denoted $A_{j,t}$. Each sector employs the same homogeneous production factors, capital $K_{j,t}$ and labor $L_{j,t}$, which are freely mobile across sectors. It follows that

$$K_{g,t} + K_{s,t} = K_t \quad \text{and} \quad L_{g,t} + L_{s,t} = L_t,$$

where K_t and L_t denote the economy's total capital and labor endowments, respectively.

Investment is produced using a CES technology

$$I_t = A_{x,t} \left(\omega^{\frac{1}{\varepsilon}} X_{g,t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)^{\frac{1}{\varepsilon}} X_{s,t}^{\frac{\varepsilon-1}{\varepsilon}} \right)^{\frac{\varepsilon}{\varepsilon-1}}, \quad (1)$$

where $X_{g,t}$ and $X_{s,t}$ denote inputs from the goods and services sectors, respectively. The parameter ε , $\varepsilon > 0$, governs the elasticity of substitution between these inputs, and ω , $\omega \in (0, 1)$, captures their relative weight in the production of investment. The term $A_{x,t}$ represents investment-specific productivity, which is neutral with respect to input composition.

Capital depreciates at a constant rate $\delta > 0$ and evolves according to the standard law of motion

$$\dot{K}_t = I_t - \delta K_t. \quad (2)$$

In equilibrium, efficiency requires that output in each sector be allocated between consumption and investment inputs such that

$$C_{g,t} + X_{g,t} = Y_{g,t} \quad \text{and} \quad C_{s,t} + X_{s,t} = Y_{s,t},$$

where $C_{g,t}$ and $C_{s,t}$ denote consumption of goods and services, respectively.

Equilibrium prices. Under constant returns to scale in the investment sector and perfectly competitive markets, the equilibrium path of relative prices depends solely on differences in sectoral technical progress. This property ensures that the evolution of prices is independent of the interaction between income and substitution effects in consumption decisions.

It is straightforward to show that at equilibrium the relative price of goods, $P_{g,t}$, to services, $P_{s,t}$, satisfies

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t}},$$

that is, it equals the inverse of the relative sectoral TFPs.

We adopt the investment good as the numeraire. It is easy to see that, at equilibrium, the prices of goods and services, relative to the prices of investment, are given by

$$P_{g,t} = \frac{\mathcal{A}_t}{A_{g,t}}, \quad \text{and} \quad P_{s,t} = \frac{\mathcal{A}_t}{A_{s,t}},$$

where

$$\mathcal{A}_t = A_{x,t} \left(\omega A_{g,t}^{\varepsilon-1} + (1-\omega) A_{s,t}^{\varepsilon-1} \right)^{\frac{1}{\varepsilon-1}}. \quad (3)$$

As shown below, $\mathcal{A}_t$ corresponds to total factor productivity in the investment sector. At equilibrium in the investment sector, the ratios of expenditure shares and input quantities on goods and services are given by

$$\frac{P_{g,t} X_{g,t}}{P_{s,t} X_{s,t}} = \frac{\omega}{1-\omega} \left(\frac{A_{s,t}}{A_{g,t}} \right)^{1-\varepsilon} \quad \text{and} \quad \frac{X_{g,t}}{X_{s,t}} = \frac{\omega}{1-\omega} \left(\frac{A_{g,t}}{A_{s,t}} \right)^{\varepsilon}.$$

In what follows, and consistently with the data, we impose the following assumption on sectoral TFP:

Assumption 1. *Sectoral total factor productivity evolves according to $A_{g,t} = A_{g,0} e^{g_g t}$, $A_{s,t} = A_{s,0} e^{g_s t}$, and $\mathcal{A}_t = \mathcal{A}_0 e^{g_A t}$, where the growth rates satisfy $g_A > g_g > g_s$.*⁵

As a consequence, the growth rates of $P_{g,t}$ and $P_{s,t}$, denoted respectively π_g and π_s , are given by

$$\pi_g = g_A - g_g < g_A - g_s = \pi_s.$$

Under Assumption 1, the model delivers the prediction, consistent with the data, that consumption service prices grow faster than consumption goods prices. In addition, the overall consumption price index increases more rapidly than the price of investment goods.

Aggregate production technology. As in the National Accounts, we define aggregate final output, measured in units of the investment good, as

$$Y_t = P_{g,t} C_{g,t} + P_{s,t} C_{s,t} + I_t. \quad (4)$$

[Herrendorf et al. \(2021\)](#) show that at equilibrium the aggregate production function is

$$Y_t = \mathcal{A}_t K_t^{\theta} L_t^{1-\theta}, \quad (5)$$

where $\mathcal{A}_t$ is the investment-sector productivity index defined in equation (3). Notice, however, that Y_t is arbitrarily measured in units of the investment good.

⁵In equation (3), $A_{x,t}$ adjusts to be consistent with this assumption.

Aggregate dynamics. Combining equations (2), (4), and (5), the law of motion for capital, for $t \geq 0$, given the exogenous path of $\mathcal{A}_t$ and an initial capital stock $K_0 > 0$, can be written as

$$\dot{K}_t = \underbrace{\mathcal{A}_t K_t^\theta L_t^{1-\theta}}_{I_t} - E_t - \delta K_t,$$

where

$$E_t = P_{g,t} C_{g,t} + P_{s,t} C_{s,t}$$

denotes total consumption expenditure, measured in units of the investment good.

Non-homothetic preferences. Let population be denoted by N_t , growing at a constant rate n , $n > 0$. At each point in time t , every individual supplies h_t units of human capital, which grows exogenously at rate g_h , $g_h > 0$. Total labor supplied is thus $L_t = h_t N_t$, and is offered inelastically.

The economy features an infinitely lived representative household whose preferences are represented by the intertemporal utility function

$$\int_0^\infty U(c_{g,t}, c_{s,t}) e^{(n-\rho)t} dt,$$

where $\rho > n$ is the subjective discount rate and $U(\cdot)$ is per capita utility. The instantaneous utility function $U(\cdot)$ is assumed to belong to the *price-independent generalized-linear* (PIGL) class –see Boppart (2014). It depends on per capita consumption of goods and services, defined as $c_{g,t} = C_{g,t}/N_t$ and $c_{s,t} = C_{s,t}/N_t$, respectively.

Since the PIGL class generally lacks a closed-form direct utility representation, we work with its indirect utility form $V(e_t, P_{g,t}, P_{s,t})$, where $e_t = E_t/N_t$ denotes per capita consumption expenditure. Following Boppart (2014), we make the following assumption

Assumption 2. *The instantaneous utility function $U(c_{g,t}, c_{s,t})$ belongs to the PIGL class and has the following indirect utility representation:*

$$V(e_t, P_{g,t}, P_{s,t}) = \frac{1}{\chi} \left(\left(\frac{e_t}{P_{s,t}} \right)^\chi - 1 \right) + \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma \right), \quad (6)$$

where $\eta > 0$ and $1 > \gamma > \chi > 0$.

The functional form of the indirect utility function in (6) implies non-homothetic preferences whenever $\chi > 0$, and it captures the two classical channels behind structural transformation: income and price effects. The two terms on the right-hand side of (6)

admit a natural interpretation. The first term represents the utility obtained from allocating all consumption expenditure to services, while the second term measures the additional utility gain from reallocating part of that expenditure to the consumption of goods. The first term depends positively on total income e , reflecting the *income effect*, while the second term depends negatively on the relative price of goods to services, P_g/P_s , reflecting the *substitution effect*.

The expenditure share on goods, derived using Roy's identity, is given by

$$s_{g,t} := \frac{P_{g,t}c_{g,t}}{e_t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma}. \quad (7)$$

The first term on the right-hand side of (7) captures the income effect: as total expenditure e_t rises, the consumption bundle shifts from goods to services when $\chi > 0$, consistent with the interpretation of services as luxuries and goods as necessities. The second term captures the substitution effect: under $\gamma > 0$, goods and services are complements, and the magnitude of the substitution effect decreases with γ .

Assumption 3 below states the necessary and sufficient condition for the indirect utility function in (6) to satisfy standard regularity conditions –Lemma 1 in [Boppart \(2014\)](#).

Assumption 3. *The following necessary and sufficient condition, for the indirect utility function in (6) to satisfy standard regularity conditions, holds*

$$e_t^{\chi} \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t}^{\gamma} P_{s,t}^{\chi-\gamma} \quad \text{or equivalently} \quad s_{g,t} \leq \frac{1-\gamma}{1-\chi}.$$

The second inequality follows directly from applying equation (7).⁶

Intertemporal problem and intratemporal allocation. Under Assumption 2, the representative household chooses a path $\{e_t, k_t\}$ for per capita consumption expenditure and capital that solves the following dynamic program

$$v(k_0) = \max_{\{e_t, k_t\}} \int_0^{\infty} V(e_t, P_{g,t}, P_{s,t}) e^{(n-\rho)t} dt, \quad (8)$$

subject to the law of motion for capital per capita

$$\dot{k}_t = \hat{\mathcal{A}}_t k_t^{\theta} - e_t - (\delta + n)k_t, \quad (9)$$

⁶It is easy to show that the left-hand-side of this condition grows faster than the right-hand-side. Consequently, the inequality holds for all $t \geq 0$ if initial conditions are such that the condition holds at time $t = 0$. In other words, the economy must be rich enough at the initial time for preferences to be well-defined.

where $k_t = K_t/N_t$ and $\hat{\mathcal{A}}_t = \mathcal{A}_t h_t^{1-\theta}$ and $\rho > n > 0$. Notice that the only relevant term in the indirect utility function (6) is $\frac{1}{\chi} \left(\frac{e_t}{P_{s,t}} \right)^\chi$. Consequently, preferences are constant intertemporal elasticity of substitution (CIES) with respect to consumption expenditure, with the discount factor given by $\Gamma_t = P_{s,t}^{-\chi} e^{(n-\rho)t}$, which from Assumption 1 declines over time since $P_{s,t}$ increases, as in the data.⁷

The Euler equation characterizing the solution to the household's problem is⁸

$$\frac{\dot{e}_t}{e_t} = \frac{1}{1-\chi} \left(\theta \hat{\mathcal{A}}_t k_t^{\theta-1} - \rho - \delta - \chi \pi_s \right). \quad (10)$$

An equilibrium path for $\{e_t, k_t\}$ is given by the system of ODEs formed by equations (9) and (10), given the initial condition $k_0, k_0 > 0$, and a standard transversality condition.⁹ The intratemporal allocation of expenditure between goods and services can then be characterized by equation (7). Given this expression, $c_{g,t}$ is determined directly. The level of $c_{s,t}$ then follows from the expenditure constraint $e_t = P_{g,t} c_{g,t} + P_{s,t} c_{s,t}$.

Aggregate Balanced Growth Path (ABGP). Along the aggregate balanced growth path (ABGP), per capita variables $\{k_t, e_t, y_t\}$ grow at the constant rate

$$g_k = \frac{g_{\mathcal{A}}}{1-\theta} + g_h.$$

From the Euler equation (10), capital per capita evolves according to

$$k_t^* = \kappa^{\frac{1}{\theta-1}} \hat{\mathcal{A}}_t^{\frac{1}{1-\theta}}, \quad \text{where} \quad \kappa := \frac{\rho + \delta + \chi \pi_s + (1-\chi)g_k}{\theta}. \quad (11)$$

Substituting into equation (9), the path of per capita consumption expenditure satisfies

$$e_t^* = (\kappa - \delta - n - g_k) k_t^*. \quad (12)$$

From the aggregate production function (5), gross and net nominal income per capita are given by

$$y_t^* = \hat{\mathcal{A}}_t k_t^{*\theta} = \kappa k_t^* \quad \text{and} \quad m_t^* = (\kappa - \delta) k_t^*,$$

where the second equality for y_t^* follows from (11). The consumption shares of gross and net income, respectively, are

$$\frac{e_t^*}{y_t^*} = \frac{\kappa - \delta - n - g_k}{\kappa} \quad \text{and} \quad \frac{e_t^*}{m_t^*} = \frac{\kappa - \delta - n - g_k}{\kappa - \delta}. \quad (13)$$

⁷In this framework the intertemporal elasticity of substitution is $1/(1-\chi)$.

⁸For a formal derivation, see Lemma 4 in Boppart (2014).

⁹In the particular case of homothetic preferences, $\chi = 0$ and the Euler equation corresponds to the case of unit intertemporal elasticity of substitution. The model becomes the two-sector growth model in Greenwood et al. (1997) without leisure.

We impose Assumption 4, under which Proposition 1 in Herrendorf et al. (2021) applies and an ABGP exists.

Assumption 4. *The discount rate satisfies*

$$\rho > \chi \left(\frac{\theta}{1-\theta} g_A + g_h + g_s \right) + n,$$

which guarantees the existence of an ABGP.

Assumption 4 is the standard condition guaranteeing that the objective in (8) is bounded.

On the Divisia index. We begin by characterizing the behavior of the Divisia index along the ABGP and decomposing its components. We then turn to its implications for welfare measurement.

From (7), the growth rate of goods consumption along the ABGP is

$$g_{c_g} = (1 - \chi) \left(\frac{\theta}{1-\theta} g_A + g_h + g_s \right) + (1 - \gamma)(g_g - g_s) > 0.$$

This growth rate is constant along the ABGP. It rises with embodied technical progress, $\frac{\theta}{1-\theta}g_A$, human capital accumulation, g_h , and technical progress in services, g_s , as these forces increase consumption expenditure measured in services units. In addition, it increases with the productivity growth differential between goods and services, with the strength of this effect governed by $1 - \gamma$.

The share of goods in total consumption expenditure is, from equation (7),

$$s_{g,t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma}.$$

Along the ABGP, $s_{g,t}$ declines over time, a well-documented feature of goods expenditure shares in the data, and converges to zero asymptotically. Consequently, if Assumption 3 holds at the initial date, it continues to hold along the ABGP.

From equation (7), real consumption of services satisfies

$$c_{s,t} = \frac{e_t}{P_{s,t}} - \eta \left(\frac{e_t}{P_{s,t}} \right)^{1-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} = (1 - s_{g,t}) \frac{e_t}{P_{s,t}}.$$

The growth rate $g_{c_s,t} := \frac{\dot{c}_{s,t}}{c_{s,t}}$ can be computed from this expression. As time goes to infinity, $g_{c_s,t}$ converges from above to

$$g_{c_s}^* = g_k - \pi_s = \frac{\theta}{1-\theta} g_A + g_h + g_s.$$

In the long run, consumption of services benefits not only from direct gains in sectoral TFP, as given by g_s , but also from gains in human capital, g_h , and investment-specific technical progress, as given by $\frac{\theta}{1-\theta}g_A$. In the transition, there is an additional effect given by sectoral reallocation (fall in $s_{g,t}$) driven by both income and substitution effects.

Finally, along the ABGP, the Divisia index for per capita gross domestic product is defined as follows:

$$g_t^D = s_e \underbrace{(s_{g,t}g_{c_g} + (1 - s_{g,t})g_{c_s,t})}_{\hat{g}_{e,t}} + (1 - s_e)g_x,$$

where $\hat{g}_{e,t}$ is the growth rate of real consumption expenditure in per capita terms, and $g_x = g_k$ is the growth rate of investment per capita. The shares of consumption expenditure in gross and net income are given by $s_e = \left\{ \frac{e_t^*}{y_t^*}, \frac{e_t^*}{m_t^*} \right\}$, respectively. From equation (13), s_e is constant along the ABGP. Moreover, $\hat{g}_{e,t}$ differs from the growth rate of consumption expenditure measured in units of the investment good, $g_e = g_k$.

Along the ABGP, output measured in units of the investment good grows at a constant rate. However, the price of consumption expenditure rises relative to the price of investment. As a result, when output is deflated by the Divisia price index, its measured growth rate declines over time.

Proposition 1. *Under Assumptions 1 to 4, along the ABGP of the structural transformation economy, the growth rate of real GDP, as measured by the Divisia index, declines over time.*

PROOF: See Appendix A.1.

The Divisia price index for consumption expenditure is given by

$$\pi_{e,t} = s_{g,t}\pi_g + (1 - s_{g,t})\pi_s.$$

Since $\pi_g > 0$, $\pi_s > 0$, and $\pi_s > \pi_g$, the relative price of investment declines over time, as observed in the data.

Moreover, this relative price decline accelerates as the share of goods in total consumption expenditure, $s_{g,t}$, decreases over time. Consequently, when real GDP is measured using the Divisia index, its growth rate—and thus measured TFP growth—declines persistently, providing a theoretical account of the observed productivity slowdown.

3 Index Number Theory and GDP Growth

This section develops a theoretical framework linking dynamic general equilibrium and index number theory, providing a rigorous foundation for measuring real income growth. We formulate the Bellman representation of household preferences and derive the associated welfare-based indices that are central to our quantitative analysis.

3.1 Bellman Representation and the Fisher–Shell Principle

Bellman representation of preferences. Following [Durán and Licandro \(2025\)](#), the Bellman representation of the preferences of the representative household at time t is given by

$$W(c_{g,t}, c_{s,t}, x_t; \nu_t) = U(c_{g,t}, c_{s,t}) + \nu_t x_t, \quad (14)$$

where $x_t = \dot{k}_t$ denotes net investment per capita and $\nu_t = v'(k_t)$ is the marginal value of capital at time t , which is taken as given when consumption and investment decisions are made. Under this representation, preferences are non-homothetic and time-varying, indexed by the marginal value of capital ν_t . The quasi-linearity of the Bellman representation arises from the additive separability of intertemporal preferences, which ensures that the marginal value of capital is independent of current consumption decisions.

To assess the welfare implications of output growth in a structural transformation economy, it is necessary to represent household preferences over per capita current consumption in goods and services and per capita current investment in a way that is compatible with intertemporal optimization. The Bellman representation offers such a framework by mapping the recursive structure of preferences into a static formulation that depends only on current choices, with net investment being valued at the marginal value of capital. This representation captures the trade-off between present consumption and future utility derived from investment, enabling meaningful welfare comparisons over time. Crucially, because it summarizes the value of postponed consumption through the marginal value of capital, the Bellman representation provides a consistent basis for applying index number theory in dynamic settings. This allows the construction of output growth indices, such as the Fisher–Shell true quantity index, that accurately reflect welfare changes without requiring knowledge of the entire sequence of consumption utility.¹⁰

¹⁰See [Durán and Licandro \(2025\)](#) for a more detailed discussion.

At any time t , the representative household maximizes the Bellman representation of preferences (14) with respect to $\{c_{g,t}, c_{s,t}, x_t\}$, subject to the per capita budget constraint

$$P_{g,t}c_{g,t} + P_{s,t}c_{s,t} + x_t = m_t, \quad (15)$$

where m_t denotes current net per capita income. Note that in equilibrium $m_t = y_t - \delta k_t$. It is important to emphasize that y_t and m_t refer to nominal income per capita, gross, and net, as they are expressed in units of the numeraire. They do not represent real expenditure, although they are measured in units of the investment good. Consequently, aggregate nominal income is given by $Y_t = y_t N_t$ and $M_t = m_t N_t$, which we take as our measures of nominal GDP and NDP, respectively. All the arguments that follow are independent of the choice of the numeraire.

Before applying index number theory to the Bellman representation in (14), Proposition 2 below derives the corresponding indirect utility function and the associated expenditure function in the particular case of PIGL preferences.

Proposition 2. *Under Assumptions 1–4, the indirect utility and expenditure functions associated with the Bellman representation of preferences in equation (14) are given by*

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^X)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu\left(m - (\nu P_s^X)^{\frac{1}{\chi-1}}\right), \quad (16)$$

and

$$f(w, P_g, P_s; \nu) = (\nu P_s^X)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^X)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}, \quad (17)$$

where w denotes a utility level as defined in (14).

PROOF: See Appendix A.2.

Thanks to the quasi-linearity of the Bellman representation, which as explained above is a direct implication of intertemporal separable preferences, the indirect utility function is linear in net income m , and the expenditure function is linear in the contribution of current consumption and current net investment to welfare w .

An important property of the HRV structural transformation model is that, in equilibrium, the marginal value of consumption expenditure measured in units of the investment good, $\frac{\partial V(\cdot)}{\partial e}$, equals the marginal value of capital, ν . This envelope condition implies that equilibrium consumption expenditure satisfies¹¹

$$e = (\nu P_s^X)^{\frac{1}{\chi-1}}. \quad (18)$$

¹¹A formal derivation is in the proof of Proposition 2 in Appendix A.2.

The higher the value of capital, ν , the greater the opportunity cost of consumption, and therefore the lower the consumption expenditure.¹²

Along the ABGP, $P_{s,t}$ and e_t grow at the constant rates $\pi_s = g_A - g_s > 0$, and $g_k = \frac{g_A}{1-\theta} + g_h$, respectively. Consequently, the marginal value of capital ν_t grows at rate $g_\nu = -((1-\chi)g_k + \chi\pi_s)$. Since $g_k > \pi_s$ and $\chi \in (0, 1)$, we have $g_k > |g_\nu| > \pi_s$, and ν_t declines more slowly than capital accumulates. Under homothetic preferences ($\chi = 0$), the income effect in equation (7) vanishes, the expression simplifies to $g_\nu = -g_k$, and the marginal value of capital declines exactly at the rate at which capital grows.

Notice that the indirect utility function (16) can be written as

$$u(m, P_g, P_s; \nu) = \nu m + \left(V \left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s \right) - \nu (\nu P_s^\chi)^{\frac{1}{\chi-1}} \right), \quad (19)$$

where νm represents the utility of allocating all income to investment, and the second term captures the welfare gain from reallocating part of it to consumption –measured as the difference between the utility of consumption and its opportunity cost. From now on, we will refer to the second term as the *consumption reallocation gain*. Similarly, we can write the expenditure function (17) as

$$f(w, P_g, P_s; \nu) = \frac{w}{\nu} - \left(\frac{V \left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s \right) - \nu (\nu P_s^\chi)^{\frac{1}{\chi-1}}}{\nu} \right), \quad (20)$$

which represents the income required to attain utility level w by investing all income, w/ν , minus the saving from reallocating part of it to consumption. The only difference between the second terms in the two expressions is the unit of measurement: in (19), it is expressed in consumption utility units; in (20), in investment units, with ν representing the marginal value of capital.

Fisher–Shell principle. Based on the Fisher–Shell principle, which states that intertemporal welfare comparisons must be made using a consistent preference ordering, and following the logic of the equivalent variation measure introduced by [Licandro et al. \(2002\)](#), and refined by [Durán and Licandro \(2025\)](#), we adopt time t as the base preference order to compare welfare at time τ with welfare at time z by means of the equivalent

¹²The initial stock of capital must be sufficiently large for ν_0 to be small enough to ensure that Assumption 3 holds.

variation measure¹³

$$\hat{m}_{t,z,\tau} = f\left(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t\right). \quad (21)$$

We use preferences ν_t to measure the income needed in τ to generate time- τ utility at z prices. In other words, the quantity $\hat{m}_{t,z,\tau}$ represents the hypothetical level of income per capita, valued at time z prices, that the representative household would have needed at time τ to attain the utility achievable under historical income and prices at τ , when evaluated using the Bellman representation of preferences at time t . In (21), the time- t Bellman representation of preferences is called the *base preference order*, and the time- z prices are called the *base prices*. They are captured by the first and second subscripts in $\hat{m}_{t,z,\tau}$, respectively. Substituting (19) evaluated at τ and (20) evaluated at z into (21), and reorganizing terms, we get

$$\hat{m}_{t,z,\tau} = m_\tau + \left(\frac{V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) - \nu_t \hat{e}_{t,\tau}}{\nu_t} \right) - \left(\frac{V(\hat{e}_{t,z}, P_{g,z}, P_{s,z}) - \nu_t \hat{e}_{t,z}}{\nu_t} \right), \quad (22)$$

where $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$, for $x = \{z, \tau\}$.¹⁴ The equivalent variation measure at τ , $\hat{m}_{t,z,\tau}$, is equal to current income, m_τ , plus the difference between consumption reallocation gains at time- τ prices and at base prices, both evaluated using time- t preferences, the base preference order. Throughout the paper, variables without a hat denote equilibrium values, whereas hatted variables represent equivalent variation measures.

The following proposition shows that the gap between the equivalent variation and the observed (equilibrium) income measures fundamentally depends on the share of goods in total consumption expenditure, evaluated at the base preference ordering. In turn, this gap is entirely driven by differences in prices between the evaluation date and the base period.

Proposition 3. *Under the conditions of Proposition 2*

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right)$$

¹³This equivalent variation measure is consistent with the equivalent variation measure suggested by Baqaee and Burstein (2022). See their Definition 3.

¹⁴When computing equivalent variation measures, $\hat{e}_{t,x}$, Assumption 3 needs to be checked for the underline preferences to be well defined.

and, for $x \in \{z, \tau\}$,

$$\hat{s}_{g,t,x} = \eta \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^{-\chi} \left(\frac{P_{g,x}}{P_{s,x}} \right)^{\gamma} \quad \text{and} \quad \hat{e}_{t,x} = (\nu_t P_{s,x}^{\chi})^{\frac{1}{\chi-1}}.$$

PROOF: See Appendix A.3.

In Proposition 3, the gap between the equivalent variation measure of income $\hat{m}_{t,z,\tau}$ and current income m_{τ} in (22) simplifies to a factor $B_{t,z,\tau}$ that multiplies the equivalent variation measure of consumption expenditure in z , $\hat{e}_{t,z}$. This factor critically depends on the difference between the equivalent variation measures of the consumption reallocation gains between τ and z , the last one corrected by the change in the price of services, the price used in (6) to measure real income gains. This deviation depends critically on the fixed-base preference order, as represented by ν_t . However, when measured around z , that is, when evaluated at $\tau = z$, it completely vanishes since consumption reallocation gains are equalized. Then $\hat{m}_{t,z,z} = m_z$ irrespective of the fixed-base preference order t .

When measuring consumption expenditure at z using the time- t Bellman representation of preferences, the equivalent variation measure values investment by ν_t instead of ν_z . Consequently, since in equilibrium the marginal value of capital decreases over time, the equivalent variation measure of consumption expenditure, $\hat{e}_{t,z}$, exceeds the observed expenditure e_z if $z < t$. At a lower investment value, it is preferable to consume more. As a result, the equivalent variation measure of the share of consumption expenditure in goods, $\hat{s}_{g,t,z}$, is lower than $s_{g,z}$ due to a stronger income effect.

It is interesting to notice that

$$\frac{m_{\tau}}{\hat{m}_{t,z,\tau}} = 1 - \frac{B_{t,z,\tau} \hat{e}_{t,z}}{\hat{m}_{t,z,\tau}}$$

is the NDP deflator associated with the per capita real NDP measure $\hat{m}_{t,z,\tau}$.

Proposition 3 is instrumental in showing the following result:

Proposition 4. *Under the conditions of Proposition 2*

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_{\tau}}{\hat{m}_{t,z,\tau}} \cdot (g_{\tau}^D + dev_{t,z,\tau}) := \hat{g}_{t,z,\tau}^D,$$

where

$$g_{\tau}^D = s_{e,\tau} (s_{g,\tau} g_{c_g,\tau} + (1 - s_{g,\tau}) g_{c_s,\tau}) + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$dev_{t,z,\tau} = \left(s_{e,\tau} s_{g,\tau} - v_{z,\tau} \check{s}_{e,t,z,\tau} \hat{s}_{g,t,\tau} \right) \pi_g + \left(s_{e,\tau} (1 - s_{g,\tau}) - v_{z,\tau} \check{s}_{e,t,z,\tau} (1 - \hat{s}_{g,t,\tau}) \right) \pi_s,$$

with

$$\check{s}_{e,t,z,\tau} := \frac{\hat{e}_{t,z}}{m_\tau} \quad \text{and} \quad v_{z,\tau} := \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}}.$$

PROOF: See Appendix A.4.

At any time τ , the growth rate of real income, as measured by the equivalent variation measure $\hat{m}_{t,z,\tau}$ in (21), is equal to the Divisia index g_τ^D plus an additive deviation $\text{dev}_{t,z,\tau}$, the sum multiplied by the scaling factor $\frac{m_\tau}{\hat{m}_{t,z,\tau}}$, which captures the discrepancy between observed income and its equivalent variation measure. Both $\text{dev}_{t,z,\tau}$ and $\frac{m_\tau}{\hat{m}_{t,z,\tau}}$ are critically influenced by the choice of base preference order and base prices. These choices significantly affect the income deflator by altering the equivalent variation measures of the expenditure share of consumption in total income, as well as the share of goods and services in total consumption expenditure. In what follows, we analyze these effects for the fixed-base Fisher–Shell (FS) and chained FS indices, along with a hybrid index: the price-chained FS index which uses a fixed preference order. Each of these indices corresponds to a different specification of the equivalent variation measures of the shares $\hat{s}_{e,t,z,\tau}$ and $\hat{s}_{g,t,z,\tau}$.

3.2 Fisher–Shell Indices

From now on, suppose that a National Statistical Agency (NSA) has recorded National-Accounts data from an initial date t_0 to the current date t_1 . The NSA seeks to measure welfare gains in a manner consistent with the Fisher–Shell principle. A key question is: for any past date $z \in (t_0, t_1)$, which *base preference order* and which *base prices* should be adopted?

To shed light on the consequences of the different approaches available, we introduce three alternative indices.

- (i) **Base-time Fisher–Shell index.** For every $z \in (t_0, t_1)$, both the base preference order and the base prices are taken from the current date t_1 ; the resulting measure is an equivalent-variation index relative to t_1 . This index is similar to the one proposed by Baqaee and Burstein (2022). For comparability, we also define the polar index, which adopts the initial period t_0 as the base. We refer to the former as the current-base FS index and to the latter as the reference-base FS index. This

is because, for comparability, we adopt t_0 as the reference time throughout the quantitative part. Because both preferences and prices are fixed, the current-base index behaves analogously to a Paasche index and the reference-base index like a Laspeyres index.

(ii) **Chained Fisher–Shell index.** At any date $z \in (t_0, t_1)$, the base preference order and base prices are those prevailing at z ; we compare z with $\tau = z - dz$ and then chain the resulting infinitesimal indices forward to obtain a chained index. This is equivalent to the chained Fisher–Shell index suggested by [Durán and Licandro \(2025\)](#).

(iii) **Price-chained Fisher–Shell index.** To isolate the roles of prices and preferences, we fix the base preference ordering at either t_1 or t_0 for all comparisons, while allowing base prices to update over time through chaining. The resulting measure is a chained price index constructed under a fixed preference ordering. If preferences are time-invariant, $\nu_t = \nu \forall t$, the index becomes a chained Fisher–Shell index.¹⁵

Comparing these three indices reveals how either dimension (preferences or prices) drives the difference between fixed-base and chained Fisher–Shell measures.

It is important to note that all Fisher–Shell indices studied in this section are welfare-based and grounded in the Fisher–Shell principle, though they rely on different base preference orders and prices. Although they yield different quantitative measures of real income, all are derived from the same underlying preference representation. The fixed-base indices apply the Fisher–Shell principle globally by evaluating welfare gains over time using a fixed preference structure and prices –either current or historical– while the chained index applies the principle locally, assessing instantaneous welfare gains at each point in time and chaining them to construct a consistent intertemporal measure. A similar idea applies to the price-chained index. As a result, they may exhibit different properties in dynamic settings.

¹⁵For completeness, we have also produced a preference-chained Fisher–Shell index, which uses fixed base prices. We fix them at t_1 or t_0 , while the preference order updates at each date z ; the resulting infinitesimal changes are then chained.

3.2.1 Chained Fisher–Shell Index

In line with [Durán and Licandro \(2025\)](#), let us first study the particular case when, for all $z \in (t_0, t_1)$, income around z is evaluated using the time- z preference order and prices. The equivalent variation measure becomes $\hat{m}_{z,z,\tau}$. From Proposition 4, when its growth rate is evaluated at $\tau = z$, $\text{dev}_{z,z,z} = 0$, implying that the growth rate at z is the Divisia index, i.e.,

$$\left. \frac{d \log \hat{m}_{z,z,\tau}}{d\tau} \right|_{\tau=z} = g_z^D.$$

When normalized to zero at $z = t_0$, the chained Fisher–Shell index becomes the chained Divisia index

$$\mathcal{D}_z = \int_{t_0}^z g_h^D dh.$$

Throughout this paper, time t_0 is said to be the *reference time*, i.e., the time at which all indices are equal to zero. The chained Divisia index is then, in the framework of the HRV structural transformation model, a welfare-based measure.

From the perspective of the chained Fisher–Shell index, the output growth rate at any point in time is computed using the preferences and prices prevailing at that moment. It captures cumulative welfare gains, as perceived by the representative agent over time, based on their evolving preferences and the changing relative prices. It is from this perspective that the chained Divisia index is welfare-based.

3.2.2 Fixed-Base Fisher–Shell Indices

Following [Baqae and Burstein \(2022\)](#), we take the current period t_1 as the base date for both preferences and prices.

In their own words, “the asymmetry of time makes current preferences more relevant than preferences in the past.” Consequently, we define what we call a *current-base Fisher–Shell index*, for all $z \in (t_0, t_1)$, as

$$\mathcal{P}_{t_1,z} = \log(\hat{m}_{t_1,t_1,z}) - \log(\hat{m}_{t_1,t_1,t_0}). \quad (23)$$

The first subscript of $\mathcal{P}$ corresponds to both the base preference order and base prices. By construction, the index is normalized at the reference time t_0 , so that $\mathcal{P}_{t_1,t_0} = 0$, with $\mathcal{P}_{t_1,z}$ measuring the cumulative growth between t_0 and z .

The value at time t_1 then satisfies

$$\mathcal{P}_{t_1, t_1} = \log(m_{t_1}) - \log(\hat{m}_{t_1, t_1, t_0}),$$

which is an equivalent variation measure of welfare gains from the initial time t_0 to the current time t_1 , evaluated using current preferences and prices. For any intermediate time $z \in (t_0, t_1)$, the difference $\mathcal{P}_{t_1, t_1} - \mathcal{P}_{t_1, z} = \log(m_{t_1}) - \log(\hat{m}_{t_1, t_1, z})$ captures the welfare gains from time z to t_1 .

The term $\hat{m}_{t_1, t_1, z}$ can be interpreted as the level of income per capita at time z , after correcting for changes in the price level, corrections being made from the perspective of the current Bellman representation of preferences and current prices. As such, the ratio $m_{t_1}/\hat{m}_{t_1, t_1, z}$ provides a money-metric measure of the welfare gains between z and t_1 when evaluated from today's point of view. This approach mirrors the logic of a Paasche index in that it values past allocations using current prices, in addition to current preferences.

We also adopt the polar view and measure welfare gains from the perspective of the initial time t_0 , our reference period. To do so, we can rely on (21) to measure welfare gains from moving from t_0 to z , for $z \in (t_0, t_1)$. This alternative approach is more closely aligned with a Laspeyres index. To fix ideas, let us adopt t_0 as base time. In what follows, we represent the reference-base Fisher–Shell index, for $z \in (t_0, t_1)$, as

$$\mathcal{L}_{t_0, z} = \log \hat{m}_{t_0, t_0, z} - \log m_{t_0}. \quad (24)$$

The index is normalized to $\mathcal{L}_{t_0, t_0} = 0$ and $\mathcal{L}_{t_0, t_1} = \log \hat{m}_{t_0, t_0, t_1} - \log m_{t_0}$.

Both the current-base and reference-base Fisher–Shell indices have associated instantaneous, welfare-based measures of growth. For any $z \in (t_0, t_1)$, the current-base growth rate is

$$\frac{d\mathcal{P}_{t_1, z}}{dz} = \frac{d \log \hat{m}_{t_1, t_1, z}}{dz} = \frac{1}{\hat{m}_{t_1, t_1, z}} \frac{d\hat{m}_{t_1, t_1, z}}{dz},$$

while the reference-base growth rate is

$$\frac{d\mathcal{L}_{t_0, z}}{dz} = \frac{d \log \hat{m}_{t_0, t_0, z}}{dz} = \frac{1}{\hat{m}_{t_0, t_0, z}} \frac{d\hat{m}_{t_0, t_0, z}}{dz}.$$

Following Durán and Licandro (2025), we establish a fundamental link between the Fisher–Shell indices and the Divisia index. As shown in Proposition 5, the instantaneous growth rate at the base time coincides with the Divisia index. Specifically, $\left. \frac{d\mathcal{P}_{t_1, z}}{dz} \right|_{z=t_1} = g_{t_1}^D$ and $\left. \frac{d\mathcal{L}_{t_0, z}}{dz} \right|_{z=t_0} = g_{t_0}^D$. It shows that the instantaneous growth rate of a base-time

Fisher–Shell index coincides with that of the Divisia index at the corresponding base time. Moreover, for any $z \in (t_0, t_1)$, the chained Fisher–Shell index lies between the current-base and reference-base Fisher–Shell indices, with the former understating and the latter overstating growth relative to the Divisia measure.

Proposition 5. *The instantaneous growth rate of the current-base and reference-base Fisher–Shell indices, when evaluated at the base time, is equal to the corresponding Divisia index:*

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D. \quad (25)$$

Moreover, for all $z \in (t_0, t_1)$, with $t_0 < t_1$, and conditional on s_{g,t_0} being small enough,

$$\frac{d\mathcal{P}_{t_1,z}}{dz} < g_z^D < \frac{d\mathcal{L}_{t_0,z}}{dz}, \quad (26)$$

and therefore

$$\mathcal{L}_{t_0,z} > \mathcal{D}_z > \mathcal{P}_{t_1,z}.$$

PROOF: See Appendix A.5.

This result implies that, while both indices match the Divisia growth rate at their respective base times, they diverge over time: the current-base Fisher–Shell index progressively underestimates growth, whereas the reference-base index overestimates it. As a consequence, when real income is repeatedly re-evaluated using a shifting current base, past economic growth appears to shrink systematically with each revision.¹⁶

3.2.3 Price-chained Fisher–Shell index

A fixed-base Fisher–Shell index evaluates income at any time $z \in (t_0, t_1)$ using base-year preferences and prices. A natural question is: how much of the difference between a chained Divisia index and a fixed-base Fisher–Shell index arises from holding prices fixed at the base year, in addition to preferences? Importantly, the principle that “current preferences are more relevant than preferences in the past” does not imply that current prices must be used when comparing two different points in the past.

¹⁶Notice that Appendix A.5 derives a sufficient condition on s_{g,t_0} . Under the calibration, $\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) < 0$, so that condition reduces to $s_{g,t_0} < \frac{\pi_s}{\pi_s - \pi_g} = 0.505$, which holds since $s_{g,80} = 0.240$. When simulating the structural transformation economy, we were unable to find any set of parameters outside this region that did not satisfy condition (26).

To isolate the role of preferences, let us construct price-chained Fisher–Shell indices that adopt either current or reference preferences as a benchmark. More formally, for all $z \in (t_0, t_1)$, define the price-chained Fisher–Shell indices as

$$\hat{\mathcal{P}}_{t_1,z} = \int_{t_0}^z \hat{g}_{t_1,h,h}^D dh, \quad \hat{\mathcal{L}}_{t_0,z} = \int_{t_0}^z \hat{g}_{t_0,h,h}^D dh,$$

where $\hat{g}_{t,h,h}^D$ is defined in Proposition 4. Any deviation from the chained Divisia index is due to the deviation term $\text{dev}_{t,z,z}$ and the ratio $m_z/\hat{m}_{t,z,z}$. The deviation reads:

$$\text{dev}_{t,z,z} = (s_{e,z}s_{g,z} - \check{s}_{e,t,z,z}\hat{s}_{g,t,z})\pi_g + (s_{e,z}(1 - s_{g,z}) - \check{s}_{e,t,z,z}(1 - \hat{s}_{g,t,z}))\pi_s,$$

where $v_{z,z} = 1$ has been used, with

$$\check{s}_{e,t,z,z} = \frac{\hat{e}_{t,z}}{m_z}.$$

In the endowment-economy version of the structural transformation model developed in the previous section, preferences are non-homothetic but time-invariant, stable in the sense of Baqaee and Burstein (2022). Consequently, applying the Fisher–Shell principle is not required. In this case, the price-chained Fisher–Shell index coincides with the chained Fisher–Shell (Divisia) index, and the current-base Fisher–Shell index collapses to a Paasche-type index, where intertemporal comparisons are made using current prices. The well-documented superiority of the chained Divisia index over the Paasche index therefore applies.

3.2.4 Summary

This section establishes the theoretical foundation for welfare-based real income measurement in an economy undergoing structural transformation. We derived the Bellman representation of preferences, which yields a static, quasi-linear structure enabling the application of index number theory. Building on this foundation, we formalized the Fisher–Shell principle and defined three key indices: the fixed-base (current- and reference-) Fisher–Shell indices, the price-chained Fisher–Shell index, and the chained Fisher–Shell index, which coincides with the chained Divisia index. Proposition 5 provides the central analytical result: while the instantaneous growth rates of the fixed-base indices match the Divisia index at their respective base times, they systematically diverge from it elsewhere. The current-base index underestimates and the reference-base index overestimates cumulative growth relative to the chained Divisia measure, with the

latter lying between them. This result quantifies the time-dependent bias inherent in fixed-base welfare comparisons and sets the stage for the quantitative evaluation in the following section.

In what follows, we quantitatively examine the behavior of these indices in the context of the HRV structural transformation model discussed above and evaluate their implications for the study of structural transformation.

4 Mapping the Model to U.S. Data

In this section, we quantitatively examine the aggregation of final expenditures on consumption goods, consumption of services, and investment using the structural transformation framework introduced in Section 2. We focus on measuring welfare gains based on the alternative indices discussed in Section 3.

To align the data from the U.S. National Accounts with the structure of the model, we follow the methodology proposed by [Herrendorf et al. \(2021\)](#). Specifically, we define goods consumption as the sum of personal consumption expenditures on goods and net exports of goods. Services consumption includes personal consumption expenditures on services, government consumption, and net exports of services. Investment is defined as the sum of private domestic investment and government investment. In the following, all references to US data use these measures.¹⁷

Our analysis proceeds in two steps. First, following [Herrendorf et al. \(2021\)](#), we extend their dataset to 2023 to document the key empirical patterns of structural transformation and evaluate the model’s ability to replicate them.¹⁸ Second, we calibrate the model to U.S. data and use it to evaluate the properties of alternative real income measures, focusing on how measurement influences the assessment of welfare gains and the interpretation of structural transformation dynamics.

¹⁷The treatment of net exports in [Herrendorf et al. \(2021\)](#) is consistent with the definition of GDP expenditure ($CGDP^e$) in the Penn World Table –see [Feenstra et al. \(2015\)](#).

¹⁸This extension uses data from the BEA National Income and Product Accounts for expenditure aggregates, the Input–Output Accounts to decompose value added into consumption and investment, the Industry Economic Accounts, and the BEA–BLS Integrated Industry-Level Production Accounts to update the March 2017 release of USA KLEMS.

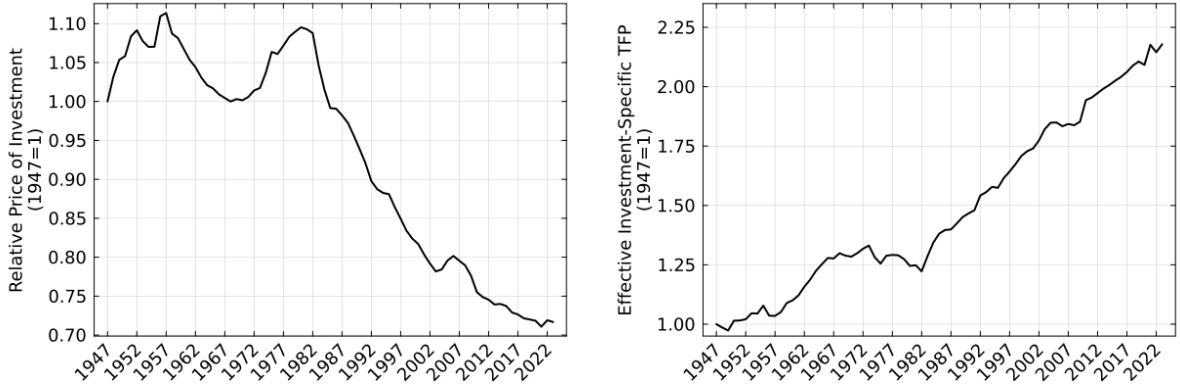


Figure 1: Relative Investment Prices and Effective Investment-Specific TFP.

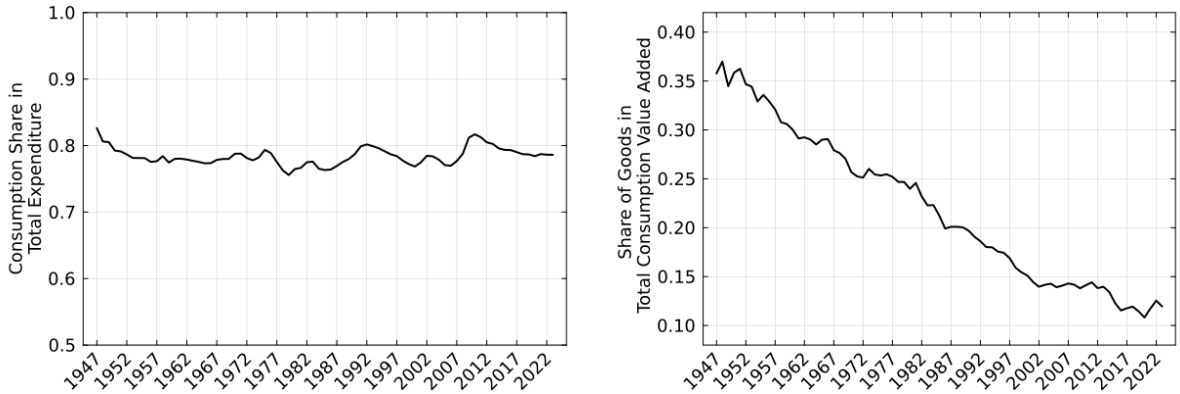


Figure 2: Consumption Expenditure and Goods Consumption Shares.

Figure 1 displays the evolution for the U.S. of the relative price of investment with respect to the price of the consumption aggregate, left panel, and the effective investment-specific TFP, right panel.¹⁹ The relative price of investment exhibits a clear downward trend beginning in 1980, falling at an average annual rate of 1.1%. This decline is mirrored by a strong upward trend in effective investment-specific TFP, which rises at an average annual rate of 1.2% over the same period.

Figure 2 complements this evidence by showing that the share of total consumption expenditure in nominal GDP fluctuates around 78% throughout the sample period. At the same time, the share of goods consumption in total nominal consumption exhibits a sustained decline, falling from around 35% in 1947 to approximately 24.0% in 1980, and

¹⁹We compute effective investment-specific TFP following the top-down approach in [Herrendorf et al. \(2021\)](#). To be precise, we apply growth accounting methods to the equilibrium aggregate production function in equation (5) and build residually a Törnqvist index for $\mathcal{A}_t$ using data on aggregate value added expressed in units of investment, aggregate capital and labor services, and average aggregate factor shares.

to 12.0% in 2023.

One challenge in mapping the model to U.S. data concerns the evolution of relative prices. While both the model and the data exhibit a declining relative price of goods to services, the rate of decline observed in the data is substantially steeper than what can be accounted for by sectoral productivity differentials alone. To reconcile this empirical discrepancy, we introduce a price wedge that affects the consumption goods sector.²⁰ Specifically, we assume that the price of consumption goods is distorted by the factor $e^{-\zeta t}$, where $\zeta > 0$ captures the additional rate of decline in the relative price of goods beyond what is explained by differences in sectoral TFP growth rates.

With this modification, the equilibrium relative price of consumption goods to consumption services becomes

$$\frac{P_{g,t}}{P_{s,t}} = \frac{A_{s,t}}{A_{g,t} e^{\zeta t}},$$

and the individual sector prices, relative to the price of investment, are given by

$$P_{g,t} = \frac{\mathcal{A}_t}{A_{g,t} e^{\zeta t}}, \quad \text{and} \quad P_{s,t} = \frac{\mathcal{A}_t}{A_{s,t}}.$$

Crucially, this wedge only affects the intratemporal allocation of consumption expenditure between goods and services, leaving the intertemporal allocation between total consumption expenditure and investment unchanged.²¹ The growth rates of consumption prices are now $\pi_g = g_A - g_g - \zeta$ and $\pi_s = g_A - g_s$, for goods and services, respectively. The overall consumption price index increases more rapidly than the price of investment goods provided that the growth rate of the wedge satisfies

$$\zeta < \frac{1}{s_g} (g_A - s_g g_g - (1 - s_g) g_s).$$

In the calibration that follows, we restrict the parameters to guarantee that $\pi_s > \pi_g$ in equilibrium, as observed in the data.

²⁰Measuring relative prices accurately is important for our quantitative exercises because structural transformation models attribute changes in sectoral expenditure shares to both income and relative price effects. If relative prices are mismeasured, the structurally estimated parameters governing these forces would be biased, distorting the inferred contribution of income and price channels to structural change. As a result, this would introduce measurement biases to our quantitative exercises.

²¹The [Online Appendix](#) explicitly incorporates this wedge into the benchmark structural transformation model of Section 2 and derives the corresponding equilibrium conditions. We model the wedge such that it does not affect production decisions, the aggregation result, or the aggregate balanced growth path. It only alters the relative prices faced by households and therefore the allocation of consumption expenditure across goods and services.

Table 1: Calibrated Parameter Values

θ	ρ	δ	n	g_h	g_A	g_s	g_g	ζ
0.333	0.04	0.08	0.0091	0.0046	0.0130	0.0024	0.0097	0.0137

Note: θ, ρ, δ are taken from HRV. TFP growth rates are computed using HRV’s sectoral TFP measures. Population growth is taken from the U.S. Census Bureau. g_h is computed using the labor services indices provided by the BEA–BLS Integrated Industry-Level Production Accounts. We estimate ζ by fitting an exponential trend to the ratio between prices of goods and services in the data versus their undistorted counterparts in the model.

4.1 Measuring GDP Growth in Practice

We assume that the distorted version of the economy described in Section 2 has been on an ABGP since 1980, and we use this restriction to discipline the measurement of GDP growth in U.S. data. We choose this starting year because, as shown in Figure 1, the relative price of investment and the effective investment-specific TFP exhibit clear trends only from 1980 onward, thus satisfying the requirement that the relative price of investment declines and that $\mathcal{A}$ grows at constant rates along the ABGP. We also assume that a National Statistical Agency (NSA) observes past and current equilibrium quantities and prices for the consumption of goods, the consumption of services, and investment, and can use this information to measure output growth. All indices reported in this section and the next are constructed in aggregate terms.

Calibration. Our strategy involves a combination of calibration and structural estimation. Parameter values are reported in Tables 1 and 2. We take the capital share parameter θ , the depreciation rate δ , and the subjective discount rate ρ directly from Herrendorf et al. (2021).

For sectoral TFPs, we assume that Assumption 1 holds and productivity grows at constant rates, such that $A_{j,t} = A_{j,80} e^{g_j t}$, for $j \in \{g, s\}$, and $\mathcal{A}_t = \mathcal{A}_{80} e^{g_A t}$. Using the same growth accounting methodology as in Herrendorf et al. (2021), we extend their series of sectoral TFPs to then normalize $A_{g,80} = A_{s,80} = \mathcal{A}_{80} = 1$, and set g_j , for $j \in \{g, s, \mathcal{A}\}$, to match the normalized TFP values at $t = 2023$.²²

²²The investment-specific TFP, $A_{x,t}$, is computed residually using equation (3) to guarantee that $\mathcal{A}_t$ grows at a constant rate. To this end, we take the parameters that discipline the sectoral weights ω and the elasticity of substitution ε in the investment aggregator (1) from Herrendorf et al. (2021).

Table 2: Preference Parameters Estimated via SMM

χ	η	γ
0.360	0.251	0.714

Note: η , χ and γ are estimated via SMM, targeting the share of consumption expenditure on goods in the data for the period of 1980–2023.

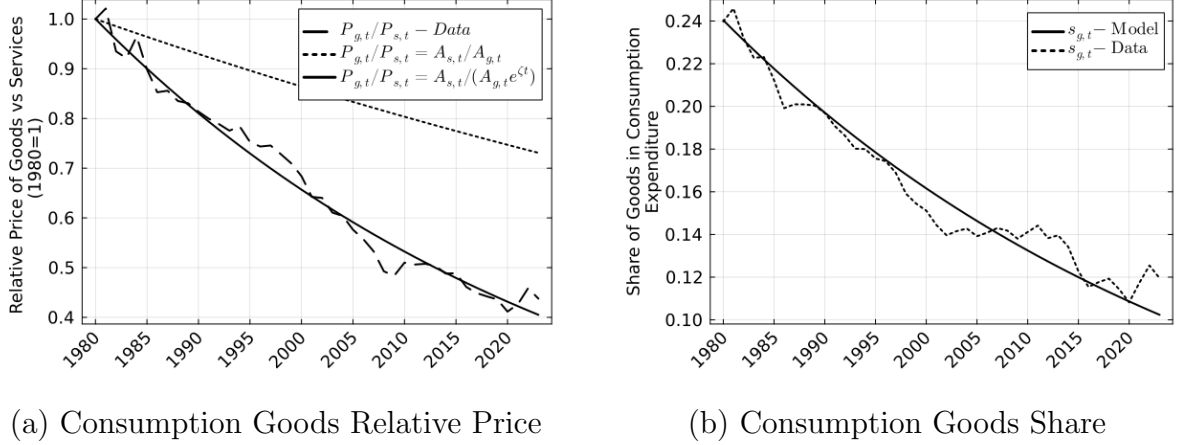


Figure 3: Consumption Goods Prices and Share

To estimate ζ , we proceed in three steps. First, we compute the relative price of consumption goods to consumption services in the data. Second, we construct the corresponding relative price implied by the model without the wedge, given by $A_{s,t}/A_{g,t}$. Third, we take the ratio of the data to the model-implied relative price and fit an exponential trend to this series. This procedure yields $\zeta = 0.0137$. Figure 3(a) shows the fit of the model together with the relative prices implied by the productivity ratio; the dotted line corresponds to the relative price without the wedge.

With these parameters in hand, we simulate the economy along the ABGP and estimate the preference parameters η, χ, γ structurally, using the Simulated Method of Moments (SMM),²³ targeting the observed share of consumption expenditure on goods from 1980 to 2023.²⁴ Figure 3(b) shows the share of goods in total consumption expen-

²³The SMM minimizes the sum of squared deviations between the model-implied and the observed shares of consumption expenditure on goods. We estimate η , χ and γ under the restrictions that $\eta > 0$ and $1 > \gamma > \chi > 0$.

²⁴In the simulated model, relative prices incorporate the wedge, ensuring that the equilibrium is consistent with the relative prices observed in the data. Importantly, the model's structure allows prices

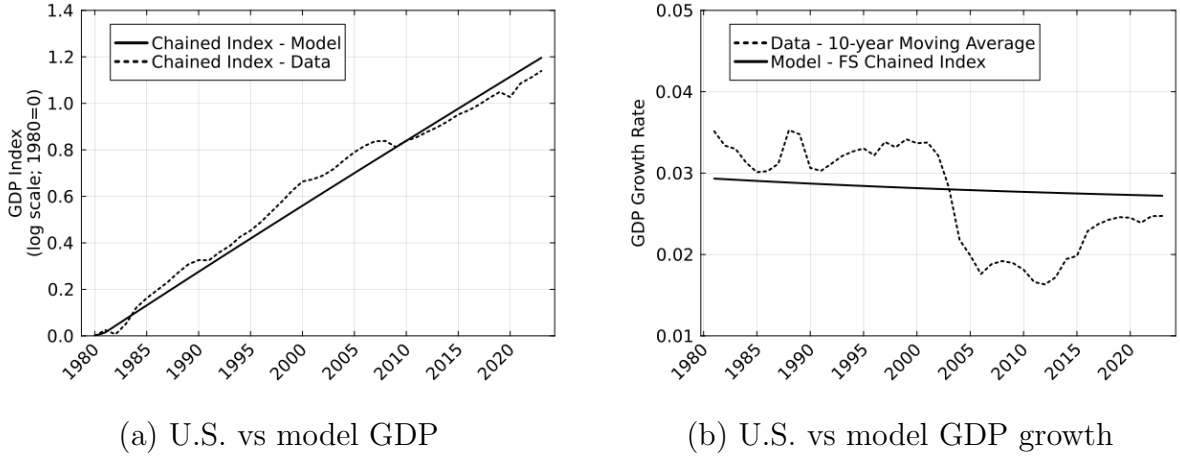


Figure 4: GDP and GDP growth

ditute in the data (dotted line) and the equilibrium share in the ABGP of the model economy (solid line).

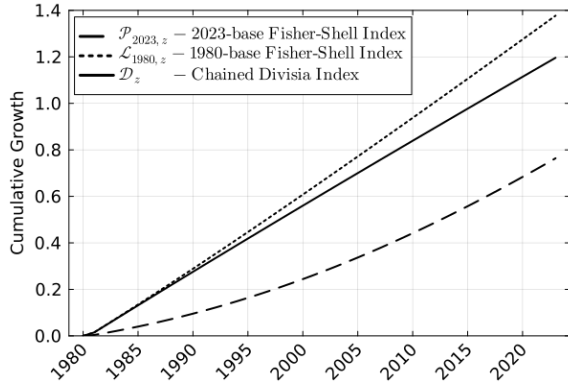
Chained Fisher–Ideal Index. Based on current and past equilibrium quantities and prices, the NSA constructs a chained Fisher–ideal index of GDP growth. In continuous time, this index coincides with the corresponding Divisia index.

Figure 4(a) plots the evolution of U.S. real GDP from 1980 to 2023. The dotted line represents the data, while the solid line shows real GDP measured using a Fisher–ideal index along the ABGP implied by the structural transformation model. The model successfully captures the underlying trend of the data. Figure 4(b) plots a 10-year moving average annual GDP growth rate in the data (dotted line) alongside the model-implied GDP growth rate over the sample period (solid line). Both series show a decline over time, consistent with the findings of [Herrendorf et al. \(2021\)](#). The slump in observed GDP growth is a direct effect of the Great Recession, which also moves the growth rate to a lower declining trend.

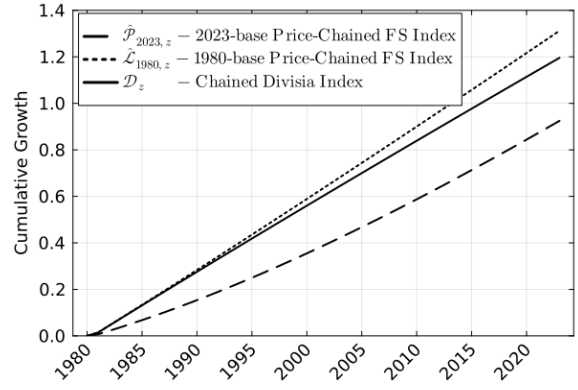
4.2 GDP Growth Measurement and Structural Transformation

In what follows, we examine the different Fisher–Shell indices introduced in the previous section and discuss their implications for measuring and interpreting structural transformation.

to be determined independently of other equilibrium objects, which are then solved conditional on these prices.



(a) Fixed-Base FS vs. Chained Divisia



(b) Price-Chained FS vs. Chained Divisia

Figure 5: Welfare-Based Real GDP Measures.

Note: Panel (a): Chained FS (solid), 1980-base FS (dotted), and 2023-base FS (dashed). –

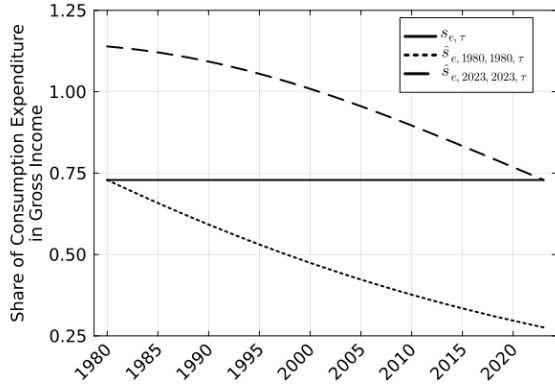
Panel (b): Chained FS (solid), 1980-base price-chained FS (dotted), and 2023-base price-chained FS (dashed).

Fixed-base vs chained Fisher–Shell indices. As part of our main discussion on welfare-based measurement of output growth, panel (a) of Figure 5 displays the evolution of real GDP (in logs) along the ABGP of the calibrated structural transformation economy. We report three alternative measures: the chained Fisher–Shell index (solid line), corresponding to the Divisia index $\mathcal{D}_t$; the 1980-base Fisher–Shell index $\mathcal{L}_{80,t}$ (dotted line); and the 2023-base Fisher–Shell index $\mathcal{P}_{23,t}$ (dashed line).

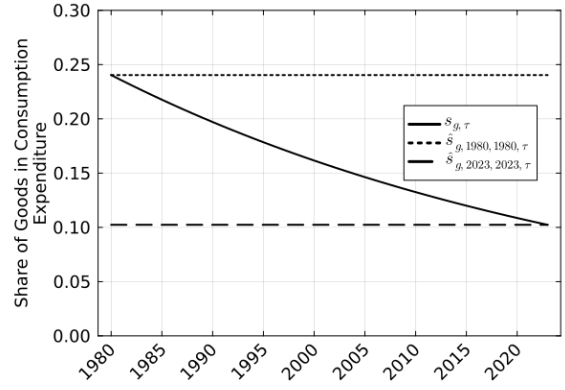
As shown in Proposition 5, relative to the chained Fisher–Shell index, the 1980-base FS index $\mathcal{L}_{80,t}$ overstates growth and the current-base FS index $\mathcal{P}_{23,t}$ understates it, with the chained FS lying between the two. Over the 1980–2023 period, the difference between the two polar indices amounts to 61.6 percentage points of cumulative real GDP growth. The fundamental reason behind these differences lies in the relative weights that each index assigns to consumption expenditure and investment when measuring the growth rate of real income.

Let us first examine the behavior of the Fisher–Shell index with a fixed base year of 1980. As shown by the dotted line in panel (a) of Figure 5, this index exceeds the chained Divisia index by 18.4 percentage points in 2023, thereby overstating income growth over the period.

For the 1980-base Fisher–Shell index, the equivalent variation measure of the con-



(a) Consumption Expenditure Share



(b) Goods Consumption Share

Figure 6: Equivalent Variation Measures for Past- and Current-base Fisher–Shell Indices.

Note: Equilibrium (solid line), 1980-base Fisher–Shell (dotted) and 2023-base Fisher–Shell (dashed). For the 1980-base index, $\hat{s}_{e,80,80,\tau}$ declines from 0.73 in 1980 to 0.28 in 2023; for the 2023-base index, it declines from approximately 1.14 in 1980 to 0.73 in 2023.

consumption expenditure share is given by –see Appendix D–

$$\hat{s}_{e,80,80,\tau} = \frac{\hat{e}_{80,80}}{\hat{m}_{80,80,\tau}} \quad \text{for all } \tau \in [1980, 2023],$$

where $\hat{e}_{80,80} = (\nu_{80} P_{s,80}^x)^{\frac{1}{x-1}}$. Since both the base preference order and the base prices are fixed at 1980, the numerator is a constant, equal to equilibrium consumption expenditure in 1980. The measure therefore varies with τ through the denominator alone. As shown in Figure 6(a), this equivalent variation measure declines with τ , which differs from the empirical evidence that nominal expenditure shares remain roughly constant over time. The decline occurs because equilibrium income m_τ , and hence the equivalent variation measure $\hat{m}_{80,80,\tau}$, increases along the ABGP while the numerator stays fixed. Adopting 1980-based preferences and 1980 prices holds hypothetical consumption expenditure at its 1980 level, so as income grows the share of consumption in the equivalent variation measure of income falls: the marginal value of investment remains constant at ν_{80} , and the household is evaluated as if it kept consuming the 1980 bundle out of a growing income.

Using past preferences reduces the willingness to consume and amplifies the incentive to invest as the economy moves away from the base year. As a result, the reference-base Fisher–Shell index assigns a growing weight to investment, which grows faster than consumption expenditure in equilibrium, causing the 1980-base FS index to rise more quickly than the Divisia index.

Figure 6(a) illustrates the behavior of the consumption expenditure share. In equilibrium, from equation (13), the consumption expenditure share remains constant at 73%, against 78% in the data –see Figure 2. In contrast, the equivalent variation measure using 1980 as the base exhibits a declining pattern: it starts at 73% in 1980 and decreases over time, reaching approximately 28% by 2023, a prediction that differs from the empirical stability of nominal consumption expenditure shares.

Notice also that when consumption expenditure at time z is evaluated using 1980 preferences, the direction of the income effect on consumption patterns reverses. Specifically, the equivalent variation measure of real consumption expenditure is given by

$$\frac{\hat{e}_{80,z}}{P_{s,z}} = (\nu_{80} P_{s,z})^{\frac{1}{\chi-1}}.$$

Since $P_{s,z}$ increases with z and $\chi - 1 < 0$, this measure of real consumption expenditure declines over time. This implies a decline in the real income effect when evaluated using 1980 preferences, leading to a shift in consumption from services to goods –see equation (7). As shown in Appendix D, this fully offsets the effect of rising service prices, which reduces the share of goods in total consumption expenditure due to complementarity in preferences. Figure 6(b) illustrates this pattern: the dotted line shows the 1980-base equivalent variation measure of the goods consumption share, $\hat{s}_{g,80,z} = 0.240$ for all z (constant at 24.0%), which is consistently higher than the corresponding equilibrium share measured under the Divisia index (which declines from 24.0% in 1980 to 10.2% in 2023).²⁵

We now turn to the 2023-base Fisher–Shell index, which evaluates real income growth using 2023 preferences and prices. As shown by the dashed line in panel (a) of Figure 5, this index substantially understates income growth relative to the chained Divisia index, leading to an index value that is 43.2 percentage points lower than the chained index in 2023.

This understatement arises for reasons symmetrical to those discussed for the 1980-base index. Specifically, for the current-base Fisher–Shell index with 2023 as the base,

²⁵Under the calibration in Table 2, $\frac{1-\gamma}{1-\chi} = 0.447$. As shown in Figure 6, the share of goods in total consumption expenditure is systematically smaller than 0.25, not only at equilibrium but also for all equivalent variation measures in both the current-base and reference-base Fisher–Shell indices. It is therefore easy to verify that Assumption 3 holds and preferences are well-defined.

the equivalent variation measure of the consumption expenditure share is given by

$$\hat{s}_{e,23,23,\tau} = \frac{\hat{e}_{23,23}}{\hat{m}_{23,23,\tau}}, \quad \text{for all } \tau \in [1980, 2023],$$

which is decreasing in τ (increasing as we move backward in time). In this case, the equivalent variation measure of consumption expenditure,

$$\hat{e}_{23,23} = (\nu_{23} P_{s,23}^\chi)^{\frac{1}{\chi-1}},$$

is again a constant, now equal to equilibrium consumption expenditure in 2023, which is higher than its 1980 counterpart. Meanwhile, equilibrium income m_τ , and hence the equivalent variation measure $\hat{m}_{23,23,\tau}$, decreases as we move back in time along the ABGP. The share therefore rises as we move into the past, again through the denominator alone. Taken together, this implies an increasing equivalent variation measure of the consumption share as we move backward in time (from 0.73 in 2023 to approximately 1.14 in 1980), which differs from the stability of nominal expenditure shares observed in the data. Thus, as we go back in time, the 2023-base Fisher–Shell index assigns a rising weight to consumption, the slowest-growing component of GDP.

As a result, the 2023-base Fisher–Shell index grows more slowly than the Divisia index and understates past economic growth and welfare gains. This behavior mirrors the distortion described for the 1980-base index but with opposite sign, as expected from the time-reversal symmetry implied by using current-base preferences –see Proposition 5. Figure 6(a) confirms this pattern: the equivalent variation measure of the consumption expenditure share is approximately 1.14 in 1980, declining over time and converging to 73% in 2023.²⁶ Figure 6(b) further illustrates that the goods consumption share for the 2023-base index, $\hat{s}_{g,23,z} = 0.102$ for all z (constant at 10.2%), is systematically below the corresponding equilibrium share (which declines from 24.0% in 1980 to 10.2% in 2023). This is explained by the reversed income effect: using future preferences to evaluate income at earlier dates generates an upward bias in real consumption expenditure and shifts consumption from goods towards services, reducing the measured share of goods in total consumption expenditure. Again, this is the mirror image of the substitution effect observed in the 1980-base case.

²⁶It is important to notice that the equivalent variation measure of investment at any time z in the past has no effect on equilibrium. It is a hypothetical calculation. As such, it is always feasible, since by taking place at a point in time, it is infinitesimally small with respect to the stock of capital.

Finally, as a robustness check, Panel (b) of Figure 5 shows the evolution of real income when measured using the price-chained Fisher Shell index at current preferences (dashed line) and 1980 preferences (dotted line). In both indices, base preferences are set at the current or reference time, but base prices are allowed to update continuously and then chained over time. Most of the discrepancy between the fixed-base Fisher–Shell indices and the Divisia index arises from holding preferences fixed over time.

Current-time upgrade and GDP history revision. If a welfare-based index of income growth is evaluated using current preferences, as suggested by Baqaee and Burstein (2022), then the entire history of economic growth is subject to continuous revision as time passes. This raises a natural question: what are the implications of such revisions for the measurement of economic growth in economies undergoing structural transformation? We address this question in the context of the structural transformation framework developed in this paper, where preferences, prices, and expenditure patterns evolve endogenously over time.

Figure 7 reports the annual growth rate of the calibrated structural transformation model, as measured by the current-base Fisher–Shell index, updated repeatedly as the base year moves forward. The chained Divisia index, shown by the solid line, indicates a declining annual growth rate of GDP between 1980 and 2023, by approximately a fifth of a percentage point. This decline reflects the slowdown in economic growth driven by structural change, as discussed in Section 4.1. In contrast, all fixed-base Fisher–Shell indices display an accelerating growth pattern: they overstate growth after the base year and understate it before. As discussed above, this discrepancy arises from evaluating income using fixed base-year preferences and prices, which generates equivalent variation expenditure shares that do not reflect the actual evolution of expenditure patterns. As a result, real income growth is systematically mismeasured over time.

Moreover, as shown in Figure 7, subsequent revisions systematically lower previous estimates of economic growth, creating the impression that earlier measures had consistently overstated growth. This pattern was a key motivation for the Bureau of Economic Analysis (BEA) to shift from fixed-base quantity indices to a chained Fisher–ideal index, underscoring the superiority of chained indices.

Summary. Using the quantitative model, we demonstrate that the paper’s main theoretical result holds in a fully specified general equilibrium environment. Specifically, the chained Fisher–Shell index yields a welfare-consistent measure of real income growth.

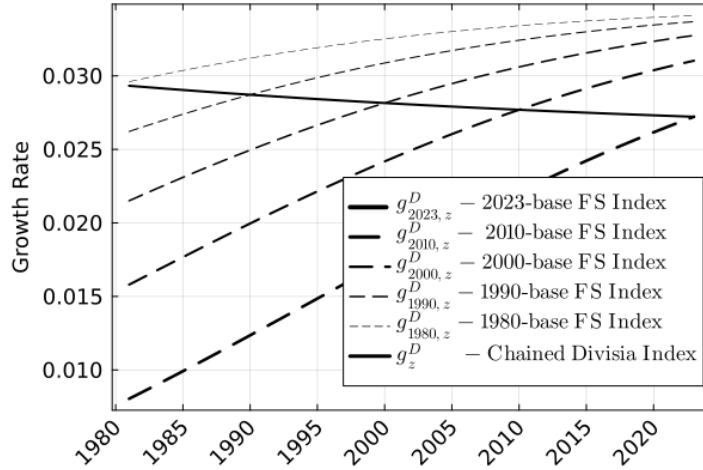


Figure 7: GDP Revisions

As formally established in Section 3.2.1, this index is equivalent to the Divisia index, which captures instantaneous welfare gains along the economy’s equilibrium path. This result also aligns the chained Fisher–Shell index with the growth rates reported in the U.S. National Accounts, which are constructed using a chained Fisher–ideal index.

In contrast, current-base Fisher–Shell indices underestimate past growth systematically. This underestimation becomes more severe the further back in time one looks, as the evaluation is conducted using today’s prices and preferences. Such undervaluation results from the behavior of expenditure shares implicit in the equivalent variation measure. Moreover, current-base indices require permanent revisions: as time passes, the base is updated, past growth is re-evaluated, and typically revised downward. This feature underscores a key limitation of fixed-base indices and highlights the advantages of chaining to capture welfare-relevant growth dynamics.²⁷

5 Non-Homothetic CES Preferences

Within the framework developed in Section 2, this section assesses whether our main results depend on the use of PIGL preferences by replacing them with the non-homothetic CES (NHCES) preferences proposed by Comin et al. (2021). The production side, the

²⁷The fixed-base problem persists even in an economy without capital, such as the purely endowment economy in Baqaee and Burstein (2022). As shown in Appendix C, the fixed-base Fisher–Shell index continues to exhibit the same qualitative distortions, still quantitatively important, when the utility function is non-homothetic and income effects are time-varying.

investment block, factor prices, and the aggregation result remain unchanged. An important feature of this specification is that the inverse of the intertemporal elasticity of substitution, given by ψ , is independent of the parameters governing the income effects across consumption categories.²⁸ The intertemporal allocation of consumption can therefore be separated from the non-homothetic reallocation of expenditure between goods and services.

Under NHCES preferences over goods and services, consumption expenditure utility C is implicitly defined by

$$\omega_c^{1/\sigma} \left(\frac{c_g}{C^{\eta_g}} \right)^{\frac{\sigma-1}{\sigma}} + (1 - \omega_c)^{1/\sigma} \left(\frac{c_s}{C^{\eta_s}} \right)^{\frac{\sigma-1}{\sigma}} = 1,$$

where $\omega_c \in (0, 1)$ determines the relative weight of goods, and $\sigma > 0$ is the elasticity of substitution between goods and services. The parameters η_g and η_s govern how the consumption of each category varies with C and hence generate non-homothetic income effects. Preferences become homothetic when $\eta_g = \eta_s$, whereas differences between these parameters induce a systematic reallocation of expenditure as aggregate consumption rises. In particular, in the development process, $\eta_s > \eta_g$ generates a gradual shift in consumption expenditure from goods towards services.

The associated expenditure function is available in closed form:

$$E(C, P_g, P_s) = \left[\omega_c P_g^{1-\sigma} C^{(1-\sigma)\eta_g} + (1 - \omega_c) P_s^{1-\sigma} C^{(1-\sigma)\eta_s} \right]^{\frac{1}{1-\sigma}},$$

where P_g and P_s denote the prices of goods and services, respectively. This expression gives the minimum expenditure required to attain a given level of composite consumption C and provides the basis for constructing the corresponding welfare-based price and quantity indices.

Calibration and model fit. Table 3 summarizes the calibration of the model with NHCES preference parameters. The parameter ψ governing the intertemporal elasticity of substitution is set to 1.5, following Buera et al. (2011), while the elasticity of substitution between goods and services, σ , is fixed at 0.2. Since σ and η_s jointly determine the evolution of sectoral expenditure shares, they cannot be separately identified from these shares alone. We therefore calibrate σ externally and estimate η_s conditional on the chosen value.

²⁸In contrast, under PIGL preferences, the inverse of the intertemporal elasticity of substitution is given by $1 - \chi$ (see equation (10)).

Table 3: Calibration of NHCES preference parameters

Parameter	Value	Target or source
ψ	1.500	Calibrated following Buera et al. (2011)
σ	0.200	Externally calibrated
ω_c	0.248	Goods expenditure share in 1980
η_g	1.000	Normalization
η_s	1.855	Evolution of consumption expenditure shares

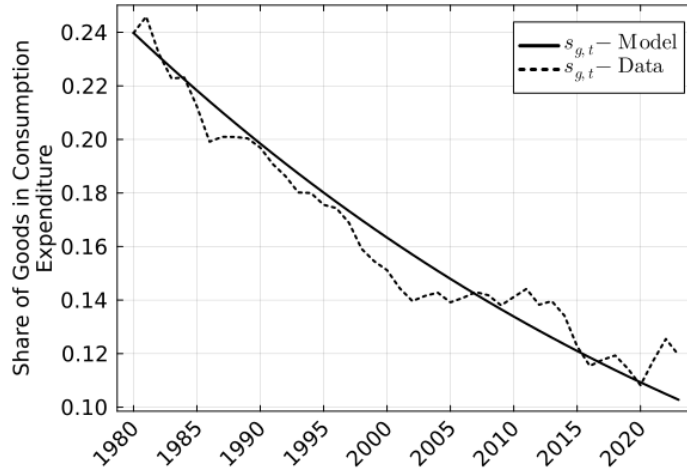


Figure 8: NHCES model: goods consumption expenditure share

The distribution parameter ω_c is set to 0.248 to match the goods expenditure share in the initial year, 1980. Since only the relative values of the income-effect parameters matter for expenditure reallocation, we normalize η_g to unity. We then estimate $\eta_s = 1.855$ by fitting the observed evolution of consumption expenditure shares. The resulting condition $\eta_s > \eta_g$ implies that services have a higher income elasticity than goods and therefore generates a gradual reallocation of consumption expenditure towards services as income rises.

Figure 8 compares the goods expenditure share generated by the model with its empirical counterpart. As under PIGL preferences, the model closely reproduces the observed decline in the goods expenditure share. Thus, replacing PIGL preferences with NHCES preferences leaves the model's ability to account for the process of structural transformation essentially unchanged.

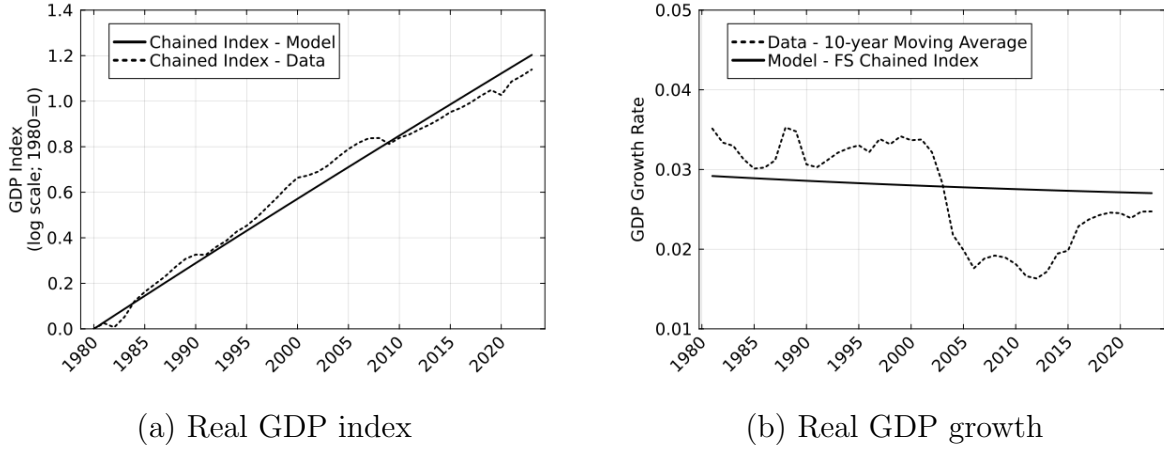


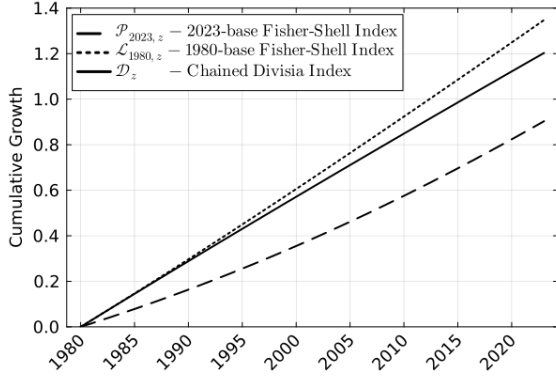
Figure 9: NHCES model: real GDP and real GDP growth

Figure 9 reports the corresponding fit for the level and growth rate of real GDP. The NHCES specification reproduces both series closely and performs as well as the baseline model with PIGL preferences. The ability of the model to account simultaneously for structural transformation and aggregate growth is therefore robust to the alternative specification of non-homothetic preferences.

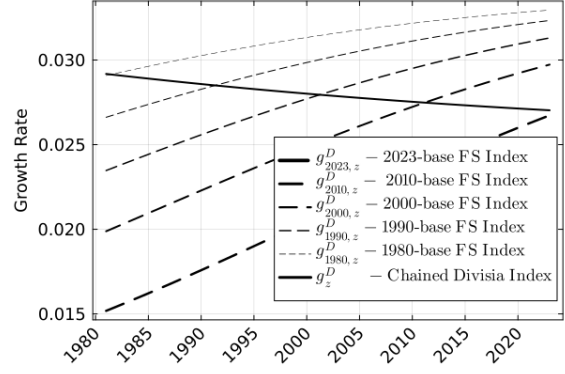
Welfare-based GDP measures. The main results are robust to replacing PIGL preferences with NHCES preferences. As shown in panel (a) of Figure 10, the same qualitative pattern emerges: the reference-base Fisher–Shell index lies above the Divisia index, whereas the current-base Fisher–Shell index lies below it. The Divisia index therefore lies between the two fixed-base welfare measures.

Panel (b) of Figure 10 reports the corresponding revisions to measured growth. For any given current base, the growth rates implied by the fixed-base indices increase over time, whereas those implied by the Divisia index decrease. Moreover, each change of base revises the fixed-base growth rates downwards relative to those previously measured. This confirms that the discrepancy between fixed-base and chained welfare measures is not a particular implication of PIGL preferences. Quantitatively, the deviations from the Divisia index under NHCES preferences are similar to those obtained under the baseline specification.

The source of these discrepancies can be understood from Figure 11. Under non-homothetic preferences, expenditure shares depend not only on relative prices but also on the level of welfare at which expenditure is evaluated. Equivalent-variation comparisons based on different reference welfare levels therefore generate different expendi-

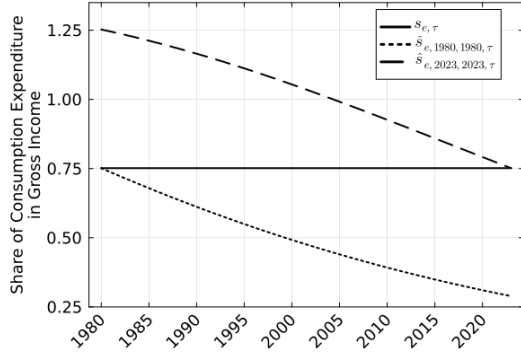


(a) Welfare-based GDP measures

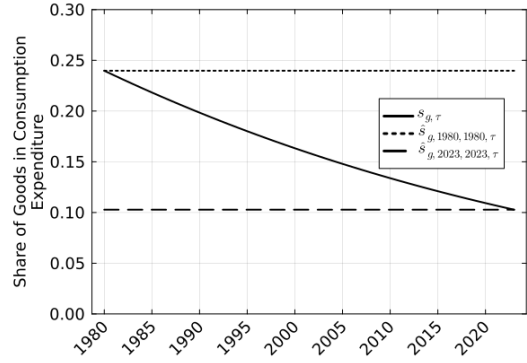


(b) Revisions to growth rates

Figure 10: NHCES model: welfare-based GDP and GDP growth measures



(a) Consumption expenditure share



(b) Goods consumption share

Figure 11: NHCES model: equivalent-variation measures of expenditure shares

ture and consumption shares. Panel (a) reports the resulting consumption expenditure shares, while panel (b) shows the corresponding goods consumption shares. The differences across reference points are the mechanism underlying the divergence between the reference-base and current-base Fisher-Shell indices. By updating the reference point in every period, the Divisia index captures these changes locally and consequently lies between the two fixed-base measures.

6 Conclusions

This paper has examined the role of real income measurement in the context of structural transformation, focusing on the welfare properties of chained indices. Within a general equilibrium model that captures key features of the U.S. structural transformation

since 1980, we compared two welfare-based approaches, the current-base Fisher–Shell equivalent variation measure inspired by [Baqaee and Burstein \(2022\)](#) and the chained Fisher–Shell index proposed by [Durán and Licandro \(2025\)](#).

Our main result is that the chained Fisher–Shell index, which updates both preferences and prices continuously, aligns with the Divisia index and provides a time-invariant, welfare-consistent measure of real GDP growth. By contrast, current-base indices distort the measurement of past growth, understating welfare gains. These distortions are particularly severe when income and substitution effects are large and relative prices evolve substantially over time, as is the case in models of structural change. These qualitative and quantitative results are robust to replacing the baseline PIGL preferences with the non-homothetic CES specification of [Comin et al. \(2021\)](#), confirming that they do not depend on the particular representation of non-homotheticity.

From a policy and statistical perspective, our findings reinforce the theoretical foundation for the use of chained indices in national accounts. While the move to chained indices was initially motivated by empirical considerations –such as reducing revisions in measured growth– their adoption also finds justification in the economic theory of index numbers when applied to dynamic, non-homothetic settings.

More broadly, this paper reaffirms the deep link between structural transformation and measurement. Accurately capturing welfare-relevant growth in the face of evolving consumption patterns requires moving beyond fixed-base approaches. Chained Fisher–Shell and Divisia indices offer flexible, theoretically grounded tools for this purpose, allowing for more accurate assessments of economic progress in transforming economies.

Appendices

A Proof of Propositions

A.1 Proof of Proposition 1

Proposition 1. *Under Assumptions 1 to 4, along the ABGP of the structural transformation economy, the growth rate of real GDP, as measured by the Divisia index, declines over time.*

PROOF: The Divisia quantity index along the ABGP of the HRV structural transformation economy is

$$g_t^D = s_e(s_{g,t} g_{c_g} + (1 - s_{g,t}) g_{c_s,t}) + (1 - s_e) g_x.$$

The corresponding Divisia price index is

$$\pi_t^D = s_e(s_{g,t} \pi_g + (1 - s_{g,t}) \pi_s).$$

Since the investment good is the numeraire, $\pi_x = 0$. It is easy to show that $g_t^D = g_k - \pi_t^D$, where g_k is the growth rate of nominal income, as measured in units of the investment good. Since $s_{g,t}$ declines along the ABGP and $\pi_s > \pi_g$, consumption inflation increases, converging to service inflation asymptotically. It follows that, the growth rate of real GDP declines over time along the ABGP. $\square$

Quantitative Evaluation. Under our estimated and calibrated parameter values, the difference in growth rates between 1980 and 2023 is

$$g_{23}^D - g_{80}^D = \pi_{80}^D - \pi_{23}^D = s_e(s_{g,80} - s_{g,23})(\pi_g - \pi_s) < 0.$$

Under our calibration, $s_e \simeq 0.73$, $s_{g,23} \simeq 0.10$, $s_{g,80} \simeq 0.24$, and $\pi_s - \pi_g \simeq 0.02$, the model implies that the annual growth rate of real GDP in 2023 is 0.21 percentage points lower than in 1980.

A.2 Proof of Proposition 2

Proposition 2. *Under Assumptions 1–4, the indirect utility and expenditure functions associated with the Bellman representation of preferences in equation (14) are, respec-*

tively, given by

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu\left(m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}\right), \quad (16)$$

and

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}, \quad (17)$$

where w is an arbitrary level of utility as in (14).

PROOF: In what follows we omit the index t to simplify notation. The household's primal problem of maximizing (14), subject to the budget constraint (15), can be solved in two stages. In the first stage, by the definition of the indirect utility function, the household solves

$$V(e, P_g, P_s) = \max_{\{c_g, c_s\}: P_g c_g + P_s c_s = e} U(c_g, c_s).$$

In the second stage, the household chooses x to solve

$$\max_x V(m - x, P_g, P_s) + \nu x.$$

The first-order condition is

$$\frac{\partial V}{\partial e}(e, P_g, P_s) = \nu.$$

Using the PIGL preferences in (6), $\frac{\partial V}{\partial e} = e^{\chi-1} P_s^{-\chi}$, we obtain

$$e^{\chi-1} P_s^{-\chi} = \nu \quad \Rightarrow \quad e = (\nu P_s^\chi)^{\frac{1}{\chi-1}}.$$

It follows that

$$x = m - e = m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}.$$

Substituting back into (6), the indirect utility function is

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu\left(m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}\right).$$

The expenditure function is derived from the dual problem

$$\min_{e, x} e + x \quad \text{s.t.} \quad V(e, P_g, P_s) + \nu x = w,$$

where w is an arbitrary level of utility. The F.O.C.s with respect to e and x are

$$1 = \mu \frac{\partial V}{\partial e} = \mu e^{\chi-1} P_s^{-\chi} \quad \text{and} \quad 1 = \mu \nu,$$

where μ is the Lagrange multiplier associated with the constraint. Equating multipliers yields

$$e^{\chi-1} P_s^{-\chi} = \nu \quad \Rightarrow \quad e = (\nu P_s^\chi)^{\frac{1}{\chi-1}}.$$

The same condition as in the primal problem. Solving for x

$$x = \frac{w - V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}.$$

Thus, the expenditure function is

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w - V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu},$$

which completes the proof. $\square$

A.3 Proof of Proposition 3.

Proposition 3. *Under the conditions of Proposition 2:*

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma}\right) \left(\frac{P_{s,z}}{P_{s,\tau}}\right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma}\right),$$

and, for $x \in \{z, \tau\}$,

$$\hat{s}_{g,t,x} = \eta \left(\frac{\hat{e}_{t,x}}{P_{s,x}}\right)^{-\chi} \left(\frac{P_{g,x}}{P_{s,x}}\right)^\gamma \quad \text{and} \quad \hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}.$$

PROOF:

Step 1. From (21),

$$\hat{m}_{t,z,\tau} = f\left(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t\right), \quad (21)$$

where $\hat{m}_{t,z,\tau}$ captures the income needed in period τ to attain the utility level of period τ under the preferences of period t and the prices of period z .

From Proposition 2,

$$u(m, P_g, P_s; \nu) = V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right) + \nu \left(m - (\nu P_s^\chi)^{\frac{1}{\chi-1}}\right), \quad (16)$$

and

$$f(w, P_g, P_s; \nu) = (\nu P_s^\chi)^{\frac{1}{\chi-1}} + \frac{w}{\nu} - \frac{V\left((\nu P_s^\chi)^{\frac{1}{\chi-1}}, P_g, P_s\right)}{\nu}. \quad (17)$$

Moreover, from Assumption 2,

$$V(e, P_g, P_s) = \frac{1}{\chi} \left(\frac{e}{P_s} \right)^\chi - \frac{\eta}{\gamma} \left(\frac{P_g}{P_s} \right)^\gamma - \frac{1}{\chi} + \frac{\eta}{\gamma}, \quad (6)$$

where $\eta > 0$ and $1 > \gamma > \chi > 0$.

Substituting (16) into (17), and the resulting expression into (21), we get

$$\begin{aligned} \hat{m}_{t,z,\tau} &= f \left(V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) + \nu_t(m_\tau - \hat{e}_{t,\tau}), P_{g,z}, P_{s,z}; \nu_t \right) \\ &= \hat{e}_{t,z} + \frac{1}{\nu_t} \left(V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) + \nu_t(m_\tau - \hat{e}_{t,\tau}) \right) - \frac{1}{\nu_t} V(\hat{e}_{t,z}, P_{g,z}, P_{s,z}), \end{aligned}$$

where $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$ for $x \in \{z, \tau\}$.

Rearranging terms, we get

$$\hat{m}_{t,z,\tau} = m_\tau + \left(\frac{V(\hat{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) - \nu_t \hat{e}_{t,\tau}}{\nu_t} \right) - \left(\frac{V(\hat{e}_{t,z}, P_{g,z}, P_{s,z}) - \nu_t \hat{e}_{t,z}}{\nu_t} \right). \quad (22)$$

Step 2. Now evaluate the numerator of both terms in brackets. Using (6), for an arbitrary $x \in \{z, \tau\}$,

$$V(\hat{e}_{t,x}, P_{g,x}, P_{s,x}) - \nu_t \hat{e}_{t,x} = \frac{1}{\chi} \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi - \frac{\eta}{\gamma} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma - \frac{1}{\chi} + \frac{\eta}{\gamma} - \nu_t \hat{e}_{t,x}.$$

From the definition $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$, we compute:

$$\frac{\hat{e}_{t,x}}{P_{s,x}} = (\nu_t P_{s,x})^{\frac{1}{\chi-1}}, \quad \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi = (\nu_t P_{s,x})^{\frac{\chi}{\chi-1}} = \nu_t \hat{e}_{t,x}.$$

Thus

$$\frac{1}{\chi} \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi - \nu_t \hat{e}_{t,x} = \left(\frac{1}{\chi} - 1 \right) \nu_t \hat{e}_{t,x}.$$

Hence:

$$V(\hat{e}_{t,x}, P_{g,x}, P_{s,x}) - \nu_t \hat{e}_{t,x} = \left(\frac{1}{\chi} - 1 \right) \nu_t \hat{e}_{t,x} - \frac{\eta}{\gamma} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma - \frac{1}{\chi} + \frac{\eta}{\gamma}.$$

Dividing by ν_t :

$$\frac{V(\hat{e}_{t,x}, P_{g,x}, P_{s,x}) - \nu_t \hat{e}_{t,x}}{\nu_t} = \left(\frac{1}{\chi} - 1 \right) \hat{e}_{t,x} - \frac{\eta}{\gamma \nu_t} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma - \frac{1}{\chi \nu_t} + \frac{\eta}{\gamma \nu_t}.$$

Substituting into (22) for $x = \tau$ and $x = z$, the constant terms $-\frac{1}{\chi \nu_t} + \frac{\eta}{\gamma \nu_t}$ cancel.

Therefore:

$$\hat{m}_{t,z,\tau} = m_\tau + \left(\frac{1}{\chi} - 1 \right) (\hat{e}_{t,\tau} - \hat{e}_{t,z}) - \frac{\eta}{\gamma \nu_t} \left(\left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma - \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \right). \quad (22')$$

Step 3. From (18), the equivalent variation measure $\hat{e}_{t,x} = (\nu_t P_{s,x}^\chi)^{\frac{1}{\chi-1}}$ implying

$$\frac{1}{\nu_t} \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^\chi = \hat{e}_{t,x}.$$

Use now Roy's identity (7) for the equivalent variation measure $\hat{s}_{g,t,x}$, $x \in \{z, \tau\}$,

$$\hat{s}_{g,t,x} = \eta \left(\frac{\hat{e}_{t,x}}{P_{s,x}} \right)^{-\chi} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma \Rightarrow \frac{\eta}{\gamma \nu_t} \left(\frac{P_{g,x}}{P_{s,x}} \right)^\gamma = \frac{1}{\gamma} \hat{s}_{g,t,x} \hat{e}_{t,x}.$$

Substituting it into (22') and using $\hat{e}_{t,\tau} = \hat{e}_{t,z} \left(\frac{P_{s,\tau}}{P_{s,z}} \right)^{\frac{\chi}{\chi-1}}$, we obtain

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right),$$

which completes the proof. $\square$

A.4 Proof of Proposition 4.

Proposition 4. *Under the conditions of Proposition 2*

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_\tau}{\hat{m}_{t,z,\tau}} \cdot (g_\tau^D + \text{dev}_{t,z,\tau}) := \hat{g}_{t,z,\tau}^D,$$

where

$$g_\tau^D = s_{e,\tau} (s_{g,\tau} g_{c_g,\tau} + (1 - s_{g,\tau}) g_{c_s,\tau}) + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$\text{dev}_{t,z,\tau} = \left(s_{e,\tau} s_{g,\tau} - v_{z,\tau} \check{s}_{e,t,z,\tau} \hat{s}_{g,t,\tau} \right) \pi_g + \left(s_{e,\tau} (1 - s_{g,\tau}) - v_{z,\tau} \check{s}_{e,t,z,\tau} (1 - \hat{s}_{g,t,\tau}) \right) \pi_s,$$

with

$$\check{s}_{e,t,z,\tau} := \frac{\hat{e}_{t,z}}{m_\tau} \quad \text{and} \quad v_{z,\tau} := \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}}.$$

PROOF:

Step 1. From Proposition 3,

$$\hat{m}_{t,z,\tau} = m_\tau + B_{t,z,\tau} \hat{e}_{t,z},$$

where

$$B_{t,z,\tau} := \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right),$$

Differentiating $B_{t,z,\tau}$ with respect to τ ,

$$\frac{dB_{t,z,\tau}}{d\tau} = -\frac{v_{z,\tau}}{\gamma} \frac{d\hat{s}_{g,t,\tau}}{d\tau} + \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \frac{dv_{z,\tau}}{d\tau},$$

where $v_{z,\tau} = \left(\frac{P_{s,z}}{P_{s,\tau}} \right)^{\frac{\chi}{1-\chi}}$. From the definition of $v_{z,\tau}$ above,

$$\frac{dv_{z,\tau}}{d\tau} = -\frac{\chi}{1-\chi} \pi_s v_{z,\tau}.$$

From the definition of $\hat{s}_{g,t,x}$ and $\hat{e}_{t,x}$, we get

$$\hat{s}_{g,t,\tau} = \eta \left(\frac{\hat{e}_{t,\tau}}{P_{s,\tau}} \right)^{-\chi} \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^{\gamma} = \eta \nu_t^{\frac{\chi}{1-\chi}} P_{s,\tau}^{\frac{\chi}{1-\chi}-\gamma} P_{g,\tau}^{\gamma}.$$

Differentiating with respect to τ ,

$$\frac{d\hat{s}_{g,t,\tau}}{d\tau} = \hat{s}_{g,t,\tau} \left[\left(\frac{\chi}{1-\chi} - \gamma \right) \pi_s + \gamma \pi_g \right].$$

Plugging both derivatives into $\frac{dB_{t,z,\tau}}{d\tau}$, we get

$$\frac{dB_{t,z,\tau}}{d\tau} = -v_{z,\tau} \left(\frac{\hat{s}_{g,t,\tau}}{\gamma} \left[\left(\frac{\chi}{1-\chi} - \gamma \right) \pi_s + \gamma \pi_g \right] + \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,\tau}}{\gamma} \right) \frac{\chi}{1-\chi} \pi_s \right),$$

which collapses to

$$\frac{dB_{t,z,\tau}}{d\tau} = -v_{z,\tau} \left(\hat{s}_{g,t,\tau} \pi_g + (1 - \hat{s}_{g,t,\tau}) \pi_s \right).$$

Step 2. The derivative of $\log \hat{m}_{t,z,\tau}$ with respect to τ is

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{\dot{m}_\tau}{\hat{m}_{t,z,\tau}} + \frac{dB_{t,z,\tau}}{d\tau} \cdot \frac{\hat{e}_{t,z}}{\hat{m}_{t,z,\tau}}.$$

Since

$$\frac{\dot{m}_\tau}{m_\tau} = s_{e,\tau} g_{e,\tau} + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$g_{e,\tau} = s_{g,\tau} (g_{c_g,\tau} + \pi_g) + (1 - s_{g,\tau}) (g_{c_s,\tau} + \pi_s),$$

Then,

$$\frac{d \log \hat{m}_{t,z,\tau}}{d\tau} = \frac{m_\tau}{\hat{m}_{t,z,\tau}} \cdot (g_\tau^D + \text{dev}_{t,z,\tau}) := \hat{g}_{t,z,\tau}^D,$$

where

$$g_\tau^D = s_{e,\tau} (s_{g,\tau} g_{c_g,\tau} + (1 - s_{g,\tau}) g_{c_s,\tau}) + (1 - s_{e,\tau}) g_{x,\tau},$$

and

$$\text{dev}_{t,z,\tau} = \left(s_{e,\tau} s_{g,\tau} - v_{z,\tau} \check{s}_{e,t,z,\tau} \hat{s}_{g,t,\tau} \right) \pi_g + \left(s_{e,\tau} (1 - s_{g,\tau}) - v_{z,\tau} \check{s}_{e,t,z,\tau} (1 - \hat{s}_{g,t,\tau}) \right) \pi_s,$$

with

$$\check{s}_{e,t,z,\tau} := \frac{\hat{e}_{t,z}}{m_\tau},$$

which completes the proof. $\square$

A.5 Proof of Proposition 5

Proposition 5. *The instantaneous growth rates of the current-base and reference-base Fisher–Shell indices, when evaluated at the base time, are equal to the corresponding Divisia index:*

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D. \quad (25)$$

Moreover, for all $z \in (t_0, t_1)$, $t_0 < t_1$, conditional on s_{g,t_0} being small enough,

$$\frac{d\mathcal{P}_{t_1,z}}{dz} < g_z^D < \frac{d\mathcal{L}_{t_0,z}}{dz}, \quad (26)$$

and therefore

$$\mathcal{L}_{t_0,z} > \mathcal{D}_z > \mathcal{P}_{t_1,z}.$$

PROOF:

Part I. Let us first prove (25). From the definition of the current-base and reference-base Fisher–Shell indices in (23) and (24), respectively,

$$\mathcal{P}_{t_1,z} = \log \hat{m}_{t_1,t_1,z} - \log \hat{m}_{t_1,t_1,t_0} \quad \text{and} \quad \mathcal{L}_{t_0,z} = \log \hat{m}_{t_0,t_0,z} - \log m_{t_0}.$$

Then, the growth rate of the current-base and reference-base Fisher–Shell indices at current and reference times, respectively, are

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = \frac{1}{\hat{m}_{t_1,t_1,z}} \cdot \left. \frac{d\hat{m}_{t_1,t_1,z}}{dz} \right|_{z=t_1} \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = \frac{1}{\hat{m}_{t_0,t_0,z}} \cdot \left. \frac{d\hat{m}_{t_0,t_0,z}}{dz} \right|_{z=t_0}.$$

From Proposition 4,

$$\left. \frac{d \log \hat{m}_{t,t,z}}{dz} \right|_{z=t} = g_t^D$$

since

$$\hat{m}_{t,t,t} = m_t \quad \text{and} \quad \text{dev}_{t,t,t} = 0.$$

The previous result is true, because $v_{t,t} = 1$, $\check{s}_{e,t,t} = s_{e,t}$ and $\hat{s}_{g,t,t} = s_{g,t}$. We have then shown that

$$\left. \frac{d\mathcal{P}_{t_1,z}}{dz} \right|_{z=t_1} = g_{t_1}^D \quad \text{and} \quad \left. \frac{d\mathcal{L}_{t_0,z}}{dz} \right|_{z=t_0} = g_{t_0}^D.$$

Part II. Let us now prove (26). From the definition of $\mathcal{P}_{t_1,z}$ and $\mathcal{L}_{t_0,z}$,

$$\frac{d\mathcal{P}_{t_1,z}}{dz} = \frac{d \log \hat{m}_{t_1,t_1,z}}{dz} \quad \text{and} \quad \frac{d\mathcal{L}_{t_0,z}}{dz} = \frac{d \log \hat{m}_{t_0,t_0,z}}{dz},$$

while from Proposition 4

$$\frac{d \log \hat{m}_{t,t,z}}{dz} = \frac{m_z}{\hat{m}_{t,t,z}} \cdot (g_z^D + \text{dev}_{t,t,z}),$$

where

$$\text{dev}_{t,t,z} = \left(s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} \right) (\pi_g - \pi_s) + \left(s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} \right) \pi_s,$$

with

$$\begin{aligned} s_{e,z} &= \frac{e_z}{m_z}, \\ s_{g,z} &= \eta \left(\frac{e_z}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^{\gamma}, \\ v_{t,z} &= \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}}, \\ \check{s}_{e,t,t,z} &= \frac{e_t}{m_z}, \\ \hat{s}_{g,t,z} &= \eta \left(\frac{\hat{e}_{t,z}}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^{\gamma}, \\ \hat{e}_{t,z} &= e_t \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{\chi-1}}. \end{aligned}$$

All of them depend exclusively on equilibrium objects.

Step 1: Let's study the factor $\frac{m_z}{\hat{m}_{t,t,z}}$. We show below that

$$\text{if } s_{g,t_0} < \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi} \pi_s - \gamma(\pi_s - \pi_g) \right)(t-z)} \quad \Rightarrow \quad \frac{m_z}{\hat{m}_{t_0,t_0,z}} > 1 > \frac{m_z}{\hat{m}_{t_1,t_1,z}}.$$

From Proposition 3

$$\hat{m}_{t,t,z} = m_z + B_{t,t,z} e_t,$$

where

$$\begin{aligned} B_{t,t,z} &= \left(\frac{1}{\chi} - 1 - \frac{\hat{s}_{g,t,z}}{\gamma} \right) \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} - \left(\frac{1}{\chi} - 1 - \frac{s_{g,t}}{\gamma} \right) \\ &= \left(\frac{1}{\chi} - 1 \right) \left(\left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} - 1 \right) + \frac{s_{g,t}}{\gamma} \left(1 - \left(\frac{P_{g,z}/P_{s,z}}{P_{g,t}/P_{s,t}} \right)^{\gamma} \right). \end{aligned}$$

The last equality is true since

$$s_{g,t} - \hat{s}_{g,t,z} \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} = s_{g,t} \left(1 - \left(\frac{P_{g,z}/P_{s,z}}{P_{g,t}/P_{s,t}} \right)^{\gamma} \right).$$

At equilibrium

$$P_{g,z} = e^{\pi_g(z-t_0)} \quad \text{and} \quad P_{s,z} = e^{\pi_s(z-t_0)},$$

with $\pi_s > 0$, $\pi_s > \pi_g$ and $1 > \gamma > \chi > 0$. Then,

$$B_{t,t,z} = \left(\frac{1}{\chi} - 1 \right) \left(e^{\frac{\chi}{1-\chi}\pi_s(t-z)} - 1 \right) + \frac{s_{g,t}}{\gamma} \left(1 - e^{\gamma(\pi_s-\pi_g)(t-z)} \right),$$

with $B_{t_1,t_1,t_1} = B_{t_0,t_0,t_0} = 0$.

Differentiate it with respect to z :

$$\frac{dB_{t,t,z}}{dz} = -\pi_s e^{\frac{\chi}{1-\chi}\pi_s(t-z)} + s_{g,t}(\pi_s - \pi_g) e^{\gamma(\pi_s-\pi_g)(t-z)}.$$

Since $\pi_s > 0$, $\pi_s > \pi_g$ and $s_{g,t} \in (0, 1)$, the first term is negative and the second is positive.

For all $z \in (t_0, t_1)$, $\frac{dB_{t_0,t_0,z}}{dz} < 0$ if and only if

$$s_{g,t_0} < \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) \right)(t_0 - z)}, \quad (s_{g,t_0})$$

Under this condition, for all $z \in (t_0, t_1)$, $B_{t_0,t_0,z} < 0$.

Notice that if $\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) < 0$, the condition becomes $s_{g,t_0} < \frac{\pi_s}{\pi_s - \pi_g}$. Otherwise $s_{g,t_0} < \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) \right)(t_0 - t_1)} < \frac{\pi_s}{\pi_s - \pi_g}$.

The consumption share of good g at time t , where $e_t = e_{t_0} e^{g_e(t-t_0)}$, is

$$s_{g,t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} = s_{g,t_0} e^{E(t-t_0)},$$

with $s_{g,t_0} = \eta e_{t_0}^{-\chi}$ and $E = -\chi(g_e - \pi_s) - \gamma(\pi_s - \pi_g)$. Given $g_e > \pi_s$, $\pi_s > \pi_g$ and $1 > \gamma > \chi > 0$, $E < 0$, so $s_{g,t}$ is strictly decreasing over time.

For all $z \in (t_0, t_1)$, $\frac{dB_{t_1,t_1,z}}{dz} < 0$ if and only if

$$s_{g,t_1} < \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) \right)(t_1 - z)}, \quad (s_{g,t_1})$$

Under this condition, for all $z \in (t_0, t_1)$, $B_{t_1,t_1,z} > 0$. Notice that this condition can be written as

$$s_{g,t_0} e^{E(t_1-t_0)} < e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) \right)(t_1-t_0)} \min_{z \in (t_0, t_1)} \frac{\pi_s}{\pi_s - \pi_g} e^{\left(\frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g) \right)(t_0 - z)},$$

with e_{t_0} determined in (12). Since $E < \frac{\chi}{1-\chi}\pi_s - \gamma(\pi_s - \pi_g)$, condition (s_{g,t_0}) implies (s_{g,t_1}) .

Step 2: Sign of $\text{dev}_{t,t,z}$. Let's study the sign of

$$\text{dev}_{t,t,z} = \left(s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} \right) (\pi_g - \pi_s) + \left(s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} \right) \pi_s.$$

With this in mind, we study first the sign of both parenthesis at the right-hand-side.

(A) Let us show that the first term in the equation above

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} > 0 \iff \nu_t > \nu_z,$$

and negative iff $\nu_t < \nu_z$. Let's prove it by steps.

Write explicitly each term at the left-hand-side of the inequality above:

$$s_{e,z} s_{g,z} = \frac{e_z}{m_z} \cdot \eta \left(\frac{e_z}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma$$

and

$$v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} \cdot \frac{e_t}{m_z} \cdot \eta \left(\frac{\hat{e}_{t,z}}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma.$$

Since, $\hat{e}_{t,z} = e_t \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{\chi-1}}$,

$$v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \frac{e_t}{m_z} \cdot \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma.$$

Take the common factor $\frac{\eta}{m_z} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma$:

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \frac{\eta}{m_z} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma (e_z^{1-\chi} P_{s,z}^\chi - e_t^{1-\chi} P_{s,t}^\chi).$$

From (18),

$$e_t = (\nu_t P_{s,t}^\chi)^{\frac{1}{\chi-1}} \implies e_t \nu_t^{\frac{-\chi}{\chi-1}} P_{s,t}^{\frac{\chi}{1-\chi}} = \nu_t^{-1}.$$

Hence, using the fact that

$$e_t^{1-\chi} P_{s,t}^\chi = \frac{1}{\nu_t} \quad \text{and} \quad e_z^{1-\chi} P_{s,z}^\chi = \frac{1}{\nu_z},$$

we obtain

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} = \frac{\eta}{m_z} \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \left(\frac{1}{\nu_z} - \frac{1}{\nu_t} \right).$$

Since $\eta > 0$, $m_z > 0$, $P_{g,z} > 0$ and $P_{s,z} > 0$, the sign is determined by $\frac{1}{\nu_z} - \frac{1}{\nu_t}$. Therefore

$$s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} > 0 \iff \frac{1}{\nu_z} > \frac{1}{\nu_t} \iff \nu_t > \nu_z,$$

and it is negative if and only if $\nu_t < \nu_z$.

Since at equilibrium $\pi_s > \pi_g$, then

$$\left(s_{e,z} s_{g,z} - v_{t,z} \check{s}_{e,t,t,z} \hat{s}_{g,t,z} \right) (\pi_g - \pi_s) < 0 \iff \frac{1}{\nu_z} > \frac{1}{\nu_t} \iff \nu_t > \nu_z,$$

and it is positive if and only if $\nu_t < \nu_z$.

(B) Let us now show that

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} > 0 \iff \nu_t > \nu_z,$$

and negative if $\nu_t < \nu_z$.

Since:

$$s_{e,z} = \frac{e_z}{m_z}, \quad v_{t,z} = \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} \quad \text{and} \quad \check{s}_{e,t,t,z} = \frac{e_t}{m_z}.$$

Thus:

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} = \frac{e_z}{m_z} - \left(\frac{P_{s,t}}{P_{s,z}} \right)^{\frac{\chi}{1-\chi}} \cdot \frac{e_t}{m_z}.$$

Take common factor $\frac{P_{s,z}^{\frac{\chi}{\chi-1}}}{m_z}$:

$$= \frac{P_{s,z}^{\frac{\chi}{\chi-1}}}{m_z} \left(e_z P_{s,z}^{\frac{\chi}{1-\chi}} - e_t P_{s,t}^{\frac{\chi}{1-\chi}} \right).$$

Since from (18)

$$e_t = (\nu_t P_{s,t}^\chi)^{\frac{1}{\chi-1}} \implies e_t P_{s,t}^{\frac{\chi}{1-\chi}} = \nu_t^{\frac{1}{\chi-1}}$$

Thus:

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} = \frac{P_{s,z}^{\frac{\chi}{\chi-1}}}{m_z} \left(\nu_z^{\frac{1}{\chi-1}} - \nu_t^{\frac{1}{\chi-1}} \right)$$

Since $\frac{1}{\chi-1} < 0$, because $\chi \in (0, 1)$, the function $x^{\frac{1}{\chi-1}}$ is decreasing in x . Hence:

$$\nu_z^{\frac{1}{\chi-1}} - \nu_t^{\frac{1}{\chi-1}} > 0 \iff \nu_z < \nu_t.$$

Consequently:

$$s_{e,z} - v_{t,z} \check{s}_{e,t,t,z} > 0 \iff \nu_t > \nu_z.$$

From the inequalities above, since $\pi_s > 0$ and $\pi_s > \pi_g$ at equilibrium, for all $z \in (t_0, t_1)$, $\text{dev}_{t_0,t_0,z} > 0$ and $\text{dev}_{t_1,t_1,z} < 0$.

Step 3. From the analysis above, for the reference-base t_0 , for all $z \in (t_0, t_1)$, $\frac{m_z}{\hat{m}_{t_0,t_0,z}} > 1$ and $\text{dev}_{t_0,t_0,z} > 0$ implying that $\frac{d\mathcal{L}_{t_0,z}}{dz} = \frac{d \log \hat{m}_{t_0,t_0,z}}{dz} > g_z^D$ and $\mathcal{L}_{t_0,z} > \mathcal{D}_z$. Using a similar argument, $\frac{d\mathcal{P}_{t_1,z}}{dz} = \frac{d \log \hat{m}_{t_1,t_1,z}}{dz} < g_z^D$ and $\mathcal{P}_{t_1,z} < \mathcal{D}_z$. $\square$

B Boppart's 2014 Lemma 1

Lemma 1 in Boppart (2014) provides sufficient conditions for the indirect utility function $V(e, P_g, P_s)$ to be well-defined. In particular, for all $t \in (t_0, t_1)$, it requires

$$e_t^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t}^\gamma P_{s,t}^{\chi-\gamma}.$$

Under the calibration presented in Section 4, this condition holds along the equilibrium path of the structural transformation economy. Using (7), the inequality above is equivalent to

$$s_{g,t} \leq \frac{1-\gamma}{1-\chi}.$$

Given the calibrated values of $\gamma = 0.714$ and $\chi = 0.360$, this bound implies that $s_{g,t} \leq 0.447$. In our simulated economy, this holds for all $t \geq 1980$ along the ABGP, as shown in Figure 6.

Moreover, for the preference order behind a current-base Fisher–Shell index $\mathcal{P}_{t_1,z}$, $z \in (t_0, t_1)$, to be well-defined,

$$\hat{e}_{t_1,z}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,z}^\gamma P_{s,z}^{\chi-\gamma} \quad \text{or equivalently} \quad \hat{s}_{g,t_1,z} \leq \frac{1-\gamma}{1-\chi}.$$

and

$$e_{t_1}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t_1}^\gamma P_{s,t_1}^{\chi-\gamma} \quad \text{or equivalently} \quad s_{g,t_1} \leq \frac{1-\gamma}{1-\chi}.$$

The first condition corresponds to the application of Boppart's Lemma 1 to the indirect utility $u(\hat{e}_{t_1,z}, P_{g,z}, P_{s,z}; \nu_{t_1})$ and the second to expenditure function $f(w, P_{g,t_1}, P_{s,t_1}; \nu_{t_1})$. The second condition holds, since Boppart's Lemma 1 holds at any point in the equilibrium path. Since ν_z is decreasing along the equilibrium path,

$$\hat{e}_{t_1,z} = \left(\frac{\nu_{t_1}}{\nu_z} \right)^{\frac{1}{\chi-1}} e_z > e_z > \frac{1-\chi}{1-\gamma} \eta P_{g,z}^\gamma P_{s,z}^{\chi-\gamma},$$

for all $z \in (t_0, t_1)$ at the ABGP equilibrium of the structural transformation economy. This can be observed in the behavior of $\hat{s}_{g,t_1,z}$ in Figure 6, which is always below the critical value 0.447.

For the preference order behind a reference-base Fisher–Shell index $\mathcal{L}_{t_0,z}$, $z \in (t_0, t_1)$, to be well-defined,

$$\hat{e}_{t_0,z}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,z}^\gamma P_{s,z}^{\chi-\gamma} \quad \text{or equivalently} \quad \hat{s}_{g,t_0,z} \leq \frac{1-\gamma}{1-\chi}.$$

and

$$e_{t_0}^\chi \geq \frac{1-\chi}{1-\gamma} \eta P_{g,t_1}^\gamma P_{s,t_1}^{\chi-\gamma}. \quad \text{or equivalently} \quad s_{g,t_0} \leq \frac{1-\gamma}{1-\chi}.$$

Figure 6 shows that $\hat{s}_{g,t_0,z}$ is also always below the critical value of 0.447, guaranteeing that both conditions are satisfied.

Hence, the regularity condition of Lemma 1 in Boppart (2014) holds along the equilibrium path. As a result, the baseline indirect utility function, as well as the preference orderings underlying the current-base and reference-base indices, are well-defined.

C Indices for Real Consumption Expenditure

This appendix applies the welfare measures discussed in this paper to an economy without capital, such as the purely endowment economy in Baqaee and Burstein (2022). As shown below, the fixed-base Fisher–Shell index continues to exhibit the same qualitative distortions –still quantitatively important– in an economy where the utility function is non-homothetic and income effects are time-varying.

The indirect utility function associated with the instantaneous utility function $U(c_{g,t}, c_{s,t})$ is

$$V(e_t, P_{g,t}, P_{s,t}) = \frac{1}{\chi} \left(\left(\frac{e_t}{P_{s,t}} \right)^\chi - 1 \right) + \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma \right). \quad (6)$$

The associated expenditure function is

$$E(v_t, P_{g,t}, P_{s,t}) = \chi^{\frac{1}{\chi}} \left(v_t + \frac{1}{\chi} - \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma \right) \right)^{\frac{1}{\chi}} P_{s,t}, \quad (27)$$

where v_t captures consumption utility level.

Let us compare welfare at time z with welfare at time t by means of the equivalent variation measure

$$\tilde{e}_{t,z} = E\left(V(e_z, P_{g,z}, P_{s,z}), P_{g,t}, P_{s,t}\right). \quad (28)$$

The representation of the instantaneous utility of consumption is time invariant. Notice that $\tilde{e}_{t,z}$ does not depend on the marginal value of capital ν_t , as is the case for $\hat{e}_{t,z}$ in (22).

Substituting (6) and (27) into (28), we get

$$\tilde{e}_{t,z} = \left(\left(\frac{e_z}{P_{s,z}} \right)^\chi + \frac{\eta\chi}{\gamma} \left(\left(\frac{P_{g,t}}{P_{s,t}} \right)^\gamma - \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \right) \right)^{\frac{1}{\chi}} P_{s,t}.$$

Chained Fisher–Shell Index. In line with [Durán and Licandro \(2025\)](#), it is easy to show that a chained Divisia index is welfare based. Differentiate (28) with respect to τ

$$\frac{d\tilde{e}_{z,\tau}}{d\tau} = \frac{dE(V_\tau, P_{g,z}, P_{s,z})}{dV_\tau} \times \frac{dV(e_\tau, P_{g,\tau}, P_{s,\tau})}{d\tau},$$

with

$$\frac{dE(V_\tau, P_{g,z}, P_{s,z})}{dV_\tau} = \chi^{\frac{1}{\chi}-1} \left(V_\tau + \frac{1}{\chi} - \frac{\eta}{\gamma} \left(1 - \left(\frac{P_{g,z}}{P_{s,z}} \right)^\gamma \right) \right)^{\frac{1}{\chi}-1} P_{s,z} = \tilde{e}_{z,\tau} \left(\frac{\tilde{e}_{z,\tau}}{P_{s,z}} \right)^{-\chi}$$

and

$$\frac{dV(e_\tau, P_{g,\tau}, P_{s,\tau})}{d\tau} = \left(\frac{e_\tau}{P_{s,\tau}} \right)^\chi (g_e - \pi_s) - \eta \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma (\pi_g - \pi_s),$$

where $g_e = \frac{\dot{e}_\tau}{e_\tau}$, $\pi_g = \frac{\dot{P}_{g,\tau}}{P_{g,\tau}}$ and $\pi_s = \frac{\dot{P}_{s,\tau}}{P_{s,\tau}}$. All three growth rates are time invariant in equilibrium. From (7)

$$\eta \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma = s_{g,\tau} \left(\frac{e_\tau}{P_{s,\tau}} \right)^\chi.$$

Then,

$$\frac{dV(e_\tau, P_{g,\tau}, P_{s,\tau})}{d\tau} = \left(\frac{e_\tau}{P_{s,\tau}} \right)^\chi (g_e - s_{g,\tau}\pi_g - (1 - s_{g,\tau})\pi_s).$$

From the definition of total consumption expenditure

$$g_e = s_{g,\tau}(g_{c_g} + \pi_g) + (1 - s_{g,\tau})(g_{c_s,\tau} + \pi_s).$$

Consequently, the growth rate at time τ when evaluated at time- z prices is

$$\frac{d\tilde{e}_{z,\tau}}{d\tau} \frac{1}{\tilde{e}_{z,\tau}} = \left(s_{g,\tau}g_{c_g} + (1 - s_{g,\tau})g_{c_s,\tau} \right) \left(\frac{e_\tau P_{s,z}}{\tilde{e}_{z,\tau} P_{s,\tau}} \right)^\chi.$$

When, for all $z \in (t_0, t_1)$, income around z is evaluated using time- z prices, the equivalent variation measure at $\tau = z$ is the Divisia index, i.e.,

$$\left. \frac{d \log \tilde{e}_{z,\tau}}{d\tau} \right|_{\tau=z} = g_{e,z}^D.$$

When normalized to zero at $z = t_0$, the chained Fisher–Shell index becomes the chained Divisia index

$$\mathcal{D}_{e,z} = \int_{t_0}^z g_{e,h}^D dh.$$

From the perspective of the chained Fisher–Shell index, the real growth rate of consumption expenditure at any point in time is computed using the prices prevailing at that moment. It captures cumulative welfare gains, as perceived by the representative agent over time. It is from this perspective that the chained Divisia index is welfare-based.

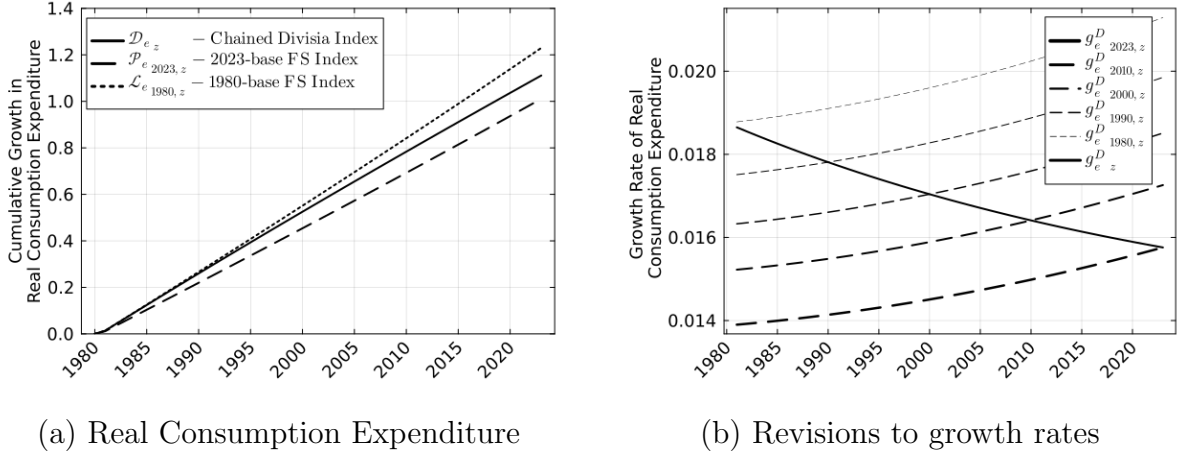


Figure 12: Real Consumption Expenditure Indices and Revisions to Growth Rates.

Note: Panel (a): Divisia (solid line), 1980-base Fisher–Shell (dotted) and 2023-base Fisher–Shell (dashed). Panel (b): Growth rates of fixed-base FS indices with bases 1980, 1990, 2000, 2010 and 2023.

Fixed-Base Fisher–Shell Indices. The current-base Fisher–Shell index is

$$\mathcal{P}_{e,t_1,z} = \log(\tilde{e}_{t_1,z}) - \log(\tilde{e}_{t_1,t_0}).$$

In turn, the reference-base Fisher–Shell index is given by

$$\mathcal{L}_{e,t_0,z} = \log(\tilde{e}_{t_0,z}) - \log(\tilde{e}_{t_0,t_0}).$$

Following the derivations above, the growth rate at z of a fixed-base Fisher–Shell index for consumption expenditure in this context, with time- t base preferences and prices is

$$g_{e,t,z} = \frac{d \log \tilde{e}_{t,z}}{dz} = g_{e,z}^D \left(\frac{e_z}{\tilde{e}_{t,z}} \frac{P_{s,t}}{P_{s,z}} \right)^\chi.$$

Quantitative results. Under the same calibration as in the main text, the resulting consumption expenditure indices are presented in Figure 12 below. As can be observed, the current-base FS index underestimates economic growth relative to the Divisia index, while the reference-base FS index overestimates it. As a result, the required permanent revision of the current-base FS index exhibits a declining growth pattern and successive downward revisions of past growth.

D Properties of the Equivalent Variation Measure

Starting from the equivalent variation measure defined in equation (21),

$$\hat{m}_{t,z,\tau} = f\left(u(m_\tau, P_{g,\tau}, P_{s,\tau}; \nu_t), P_{g,z}, P_{s,z}; \nu_t\right),$$

we now examine its structure and implications. This equivalent variation results from solving two consecutive problems.

First stage. At the first stage, the representative household uses time- t preferences to evaluate the utility attainable at time- τ with income m_τ and prices $P_{g,\tau}$ and $P_{s,\tau}$. To do so, the household solves the following hypothetical problem:

$$\max_{\tilde{c}_g, \tilde{c}_s, \tilde{x}} U(\tilde{c}_{g,t,\tau}, \tilde{c}_{s,t,\tau}) + \nu_t \tilde{x}_{t,\tau}$$

subject to

$$m_\tau = P_{g,\tau} \tilde{c}_{g,t,\tau} + P_{s,\tau} \tilde{c}_{s,t,\tau} + \tilde{x}_{t,\tau}.$$

Using the indirect utility function (6), this problem reduces to:

$$\max_{\tilde{e}, \tilde{x}} V(\tilde{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau}) + \nu_t \tilde{x}_{t,\tau}$$

subject to

$$m_\tau = \tilde{e}_{t,\tau} + \tilde{x}_{t,\tau},$$

where $\tilde{e}_{t,\tau} = P_{g,\tau} \tilde{c}_{g,t,\tau} + P_{s,\tau} \tilde{c}_{s,t,\tau}$.

The first-order condition with respect to $\tilde{e}_{t,\tau}$ yields, after some algebra,

$$\tilde{e}_{t,\tau} = (\nu_t P_{s,\tau}^\chi)^{\frac{1}{\chi-1}}.$$

Since the price of services grows with τ , the hypothetical consumption expenditure $\tilde{e}_{t,\tau}$ is decreasing in τ . It is important to note, however, that e_τ is increasing over time, because ν_τ declines at a faster rate than the price of services increases. The hypothetical consumption expenditure share is

$$\tilde{s}_{e,t,\tau} = \frac{\tilde{e}_{t,\tau}}{m_\tau}.$$

It is decreasing with τ since $\tilde{e}_{t,\tau}$ is decreasing and m_τ increasing.

Applying Roy's identity to $V(\tilde{e}_{t,\tau}, P_{g,\tau}, P_{s,\tau})$, the hypothetical share of goods on total consumption expenditure is

$$\tilde{s}_{g,t,\tau} := \frac{P_{g,\tau} \tilde{c}_{g,t,\tau}}{\tilde{e}_{t,\tau}} = \eta \left(\frac{\tilde{e}_{t,\tau}}{P_{s,\tau}} \right)^{-\chi} \left(\frac{P_{g,\tau}}{P_{s,\tau}} \right)^\gamma.$$

The hypothetical share of goods in total consumption may be growing or declining. Since $P_{g,\tau}/P_{s,\tau}$ is declining in τ and $\gamma > 0$, the relative price effect unambiguously reduces $\tilde{s}_{g,t,\tau}$. The income effect works in the opposite direction: $\tilde{e}_{t,\tau}/P_{s,\tau}$ is declining in τ and, with $\chi > 0$, this raises $(\tilde{e}_{t,\tau}/P_{s,\tau})^{-\chi}$, increasing the share of goods. This is the reverse of the standard case where both preferences and prices vary over time, in which rising income drives expenditure toward services.

Second stage. At the second stage, $\hat{m}_{t,z,\tau}$ is determined by equation (21). By construction, when using time- t preferences, as represented by ν_t , there exists a vector $(\hat{c}_{g,t,z,\tau}, \hat{c}_{s,t,z,\tau}, \hat{x}_{t,z,\tau})$ that solves the maximization problem:

$$\max_{\hat{c}_g, \hat{c}_s, \hat{x}} U(\hat{c}_{g,t,z,\tau}, \hat{c}_{s,t,z,\tau}) + \nu_t \hat{x}_{t,z,\tau}$$

subject to

$$\hat{m}_{t,z,\tau} = P_{g,z} \hat{c}_{g,t,z,\tau} + P_{s,z} \hat{c}_{s,t,z,\tau} + \hat{x}_{t,z,\tau}.$$

Using the indirect utility function (6), this problem reduces to:

$$\max_{\hat{e}, \hat{x}} V(\hat{e}_{t,z,\tau}, P_{g,z}, P_{s,z}) + \nu_t \hat{x}_{t,z,\tau}$$

subject to

$$\hat{m}_{t,z,\tau} = \hat{e}_{t,z,\tau} + \hat{x}_{t,z,\tau},$$

where $\hat{e}_{t,z,\tau} = P_{g,z} \hat{c}_{g,t,z,\tau} + P_{s,z} \hat{c}_{s,t,z,\tau}$.

The first-order condition with respect to $\hat{e}_{t,z,\tau}$ yields, after some algebra,

$$\hat{e}_{t,z,\tau} = (\nu_t P_{s,z}^\chi)^{\frac{1}{\chi-1}} := \hat{e}_{t,z}.$$

Notice that $\tilde{e}_{t,\tau}$ and $\hat{e}_{t,z}$ are formally identical.

Since the Bellman representation is quasilinear in $\hat{x}$, optimal consumption expenditure $\hat{e}$ does not depend on $\hat{m}$ and thus becomes independent of time τ . In other words, the household optimally chooses a constant hypothetical consumption expenditure $\hat{e}_{t,z}$, for all τ throughout the evaluated period, which depends only on the base preference parameter ν_t and base prices $P_{s,z}$.

The equivalent variation measure of the consumption expenditure share is

$$\hat{s}_{e,t,z,\tau} = \frac{\hat{e}_{t,z}}{\hat{m}_{t,z,\tau}}.$$

As long as hypothetical income grows with τ , the hypothetical consumption expenditure share is decreasing.

Applying Roy's identity to $V(\hat{e}_{t,z}, P_{g,z}, P_{s,z})$, which is independent of τ , gives the equivalent variation measure of the goods consumption share:

$$\hat{s}_{g,t,z} := \frac{P_{g,z} \hat{c}_{g,t,z}}{\hat{e}_{t,z}} = \eta \left(\frac{\hat{e}_{t,z}}{P_{s,z}} \right)^{-\chi} \left(\frac{P_{g,z}}{P_{s,z}} \right)^{\gamma},$$

the same for all τ throughout the evaluated period. Thus, the equivalent variation measure of the share of goods in total consumption is constant for all τ . Note also that $\hat{s}_{g,t,z}$ and $\tilde{s}_{g,t,\tau}$ are formally identical.

The constancy of $\hat{s}_{g,t,z}$ follows from two mutually reinforcing forces. First, the income effect is muted: quasilinearity pins down optimal consumption expenditure $\hat{e}_{t,z}$ independently of τ , so there is no income variation to shift the goods share. Second, the relative price effect is also muted: since prices are fixed at z , the term $(P_{g,z}/P_{s,z})^{\gamma}$ is constant throughout the evaluated period. With both channels shut down, $\hat{s}_{g,t,z}$ is invariant to τ .

Current-base Fisher–Shell index. For a fixed-base Fisher–Shell index, with base t , for all $\tau \in (t_0, t_1)$,

$$\hat{s}_{e,t,t,\tau} = \frac{e_t}{\hat{m}_{t,t,\tau}} \quad \text{and} \quad \hat{s}_{g,t,t} = \eta \left(\frac{e_t}{P_{s,t}} \right)^{-\chi} \left(\frac{P_{g,t}}{P_{s,t}} \right)^{\gamma} = s_{g,t}.$$

Then, for the current-base FS index the equivalent variation measure of the share of goods consumption is constant, equal to the share in current period s_{g,t_1} . The consumption expenditure share is declining, since e_{t_1} is constant but $\hat{m}_{t_1,t_1,\tau}$ is increasing with τ , converging from above to s_{e,t_1} as time converges to t_1 .

Replication Files and Online Appendix

Supplementary replication materials and the Online Appendix can be found in the paper's GitHub Repository at: <https://github.com/jvizcaino/Chained-Indices-Unchained>.

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